Multivariate Real Analysis

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1 Differentiation

1.1 Partial, Directional, and Path Derivatives

A simple way to extract derivatives is to fix all of the input components except for one, and treat f as a single-variable function. This results in—for now—a completely separate notion of a derivative.

Definition 1.1 (Partial Derivative)

Given function $f : E \subset \mathbb{R}^n \to \mathbb{R}^m$, the ith **partial derivative** of f at $a \in E$ is defined

$$\frac{\partial f}{\partial x_i}\bigg|_a := \lim_{h \to 0} \frac{f(a + he_i)}{h} \tag{2}$$

where e_i is the ith basis vector of $\mathbb{R}^n$.

Since we are fixing every other parameter except for the ith one, we can find the partial derivative by differentiating the function w.r.t. x_i and fixing all other variables.

Example 1.1 (Partial Derivatives of Saddle Points)

To find all partial derivatives of the function

$$f(x, y) = x^2 - y^2 \tag{3}$$

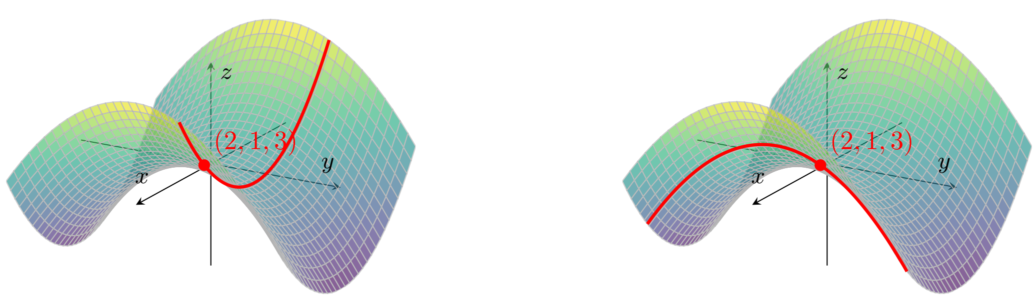
We compute

$$\frac{\partial f}{\partial x} = 2x \tag{4}$$

$$\frac{\partial f}{\partial y} = -2y. \tag{5}$$

At the point $(2, 1)$, these become

$$\frac{\partial f}{\partial x}(2, 1) = 4 \qquad \text{and} \qquad \frac{\partial f}{\partial y}(2, 1) = -2. \tag{6}$$



(a) Slicing the saddle surface along the plane $y = 1$, the resulting curve has slope 4 at $(2, 1, 3)$.

(b) Slicing along the plane $x = 2$ gives a curve with slope -2 at the same point.

Figure 1: Two cross-sections of the saddle surface $z = x^2 - y^2$ through the point $(2, 1, 3)$. Each slice isolates one partial derivative by holding the other variable fixed.

Example 1.2 (Partial Derivatives Using Differentiation Rules)

To find all partial derivatives of the function

$$f(x, y, z) = x \ln(y^2 + z^2), \tag{7}$$

We compute

$$\frac{\partial f}{\partial x} = \ln(y^2 + z^2) \tag{8}$$
$$\frac{\partial f}{\partial y} = x \cdot \frac{1}{y^2 + z^2} \cdot (2y) = \frac{2xy}{y^2 + z^2} \tag{9}$$
$$\frac{\partial f}{\partial z} = x \cdot \frac{1}{y^2 + z^2} \cdot (2z) = \frac{2xz}{y^2 + z^2} \tag{10}$$

Example 1.3 (Partial Derivatives Using Limit Definition)

Let

$$f(x, y) = \begin{cases} \dfrac{xy}{x^2 + y^2} & (x, y) \neq (0, 0), \\ 0 & (x, y) = (0, 0). \end{cases} \tag{11}$$

We compute the partial derivatives $f_x(0, 0)$ and $f_y(0, 0)$ using the limit definition.
First, for the partial derivative with respect to x, we have

$$\begin{aligned} f_x(0, 0) &= \lim_{h \to 0} \frac{f(h, 0) - f(0, 0)}{h} && (12) \\ &= \lim_{h \to 0} \frac{\frac{h(0)}{h^2 + 0^2} - 0}{h} && (13) \\ &= \lim_{h \to 0} \frac{0}{h} && (14) \\ &= 0. && (15) \end{aligned}$$

Similarly, for the partial derivative with respect to y, we have

$$\begin{aligned} f_y(0, 0) &= \lim_{h \to 0} \frac{f(0, h) - f(0, 0)}{h} && (16) \\ &= \lim_{h \to 0} \frac{\frac{0(h)}{0^2 + h^2} - 0}{h} && (17) \\ &= \lim_{h \to 0} \frac{0}{h} && (18) \\ &= 0. && (19) \end{aligned}$$

Therefore,

$$f_x(0, 0) = 0 \qquad \text{and} \qquad f_y(0, 0) = 0. \tag{20}$$

Note that even though the limit of $f(x, y)$ as $(x, y) \to (0, 0)$ does not exist, the partial derivatives at $(0, 0)$ still exist.

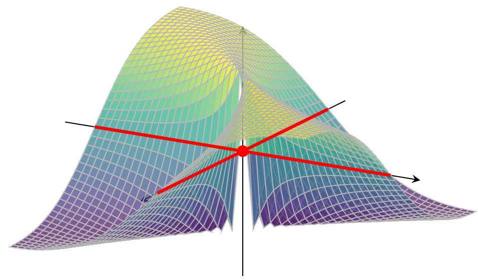


Figure 2: Both coordinate cross-sections of $f(x,y) = \frac{xy}{x^2+y^2}$ through the origin lie on the line $z = 0$, which shows why $f_x(0,0) = 0$ and $f_y(0,0) = 0$ even though the full two-variable limit does not exist.

Theorem 1.1 (Bounded Partials Implies Continuity)

If f is a real-valued function defined in an open set $E \subset \mathbb{R}^n$ and the partial derivatives $\partial_1 f, \ldots, \partial_n f$ are bounded in E, then f is continuous in E.

Proof. Since the partials exist, this means that the cross-sections are differentiable, so we take a stepwise approach. Let the partials be bounded by M. Fix $\epsilon > 0$ and $x \in E$. Then, for $y \in E$, let $\phi_k = (x_1, \ldots, x_k, y_{k+1}, \ldots, y_n)^T$. So,

$$\|f(x) - f(y)\| = \left\| \sum_{k=1}^n f(\phi_k(x)) - f(\phi_{k-1}(x)) \right\| \tag{21}$$

$$\leq \sum_{k=1}^n \|f(\phi_k(x)) - f(\phi_{k-1}(x))\| \tag{22}$$

$$\leq \sum_{k=1}^n M|y_k - x_k| \tag{23}$$

So setting $\delta = \frac{\epsilon}{nM}$ gives the desired result.

The partial derivative looks at the function as it is approaching a along an axis, but we can consider any direction in the domain.

Definition 1.2 (Directional Derivative)

Given function $f : E \subset \mathbb{R}^n \to \mathbb{R}^m$, the **directional derivative** of f at $a \in E$ in direction $u \in \mathbb{R}^n$ is defined

$$D_v f_a \coloneqq \lim_{h \to 0} \frac{f(a + hu) - f(a)}{h} \tag{24}$$

It should be obvious that if all directional derivatives exist, then the partials exist (since we can just set the directional vectors to be the unit vectors).

When computing directional derivatives, it is convenient to normalize the directional vector v to be unit length so that it coincides with the partial derivatives. We don't technically need to set $\|v\| = 1$, but if we have two vectors v and a scaled cv, then the directional derivatives will also be scaled ($\nabla_{cv} f(a) = c\nabla_v f(a)$), so we will only work with unit directional vectors. Some say that this restriction is undesirable, since it loses the linearity of the function $v \mapsto \partial_v f(a)$.

Example 1.4

Let

$$f(x, y) = x^2 y + 4y^2. \tag{25}$$

We compute the directional derivative of f at the point $(2, 1)$ in the direction of the vector $v = (1, 1)$. After normalizing, we have $u = \frac{v}{\|v\|} = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$. By the limit definition of the directional derivative,

$$\begin{aligned} D_u f(2, 1) &= \lim_{h \to 0} \frac{f\big((2, 1) + hu\big) - f(2, 1)}{h} && (26) \\ &= \lim_{h \to 0} \frac{f\left(2 + \frac{h}{\sqrt{2}}, 1 + \frac{h}{\sqrt{2}}\right) - f(2, 1)}{h}. && (27) \end{aligned}$$

Now,

$$f(2, 1) = 2^2(1) + 4(1)^2 = 8. \tag{28}$$

Also,

$$\begin{aligned} f\left(2 + \frac{h}{\sqrt{2}}, 1 + \frac{h}{\sqrt{2}}\right) &= \left(2 + \frac{h}{\sqrt{2}}\right)^2 \left(1 + \frac{h}{\sqrt{2}}\right) + 4\left(1 + \frac{h}{\sqrt{2}}\right)^2 && (29) \\ &= 8 + \frac{16h}{\sqrt{2}} + \frac{9h^2}{2} + \frac{h^3}{2\sqrt{2}}. && (30) \end{aligned}$$

Therefore,

$$\begin{aligned} D_u f(2, 1) &= \lim_{h \to 0} \frac{8 + \frac{16h}{\sqrt{2}} + \frac{9h^2}{2} + \frac{h^3}{2\sqrt{2}} - 8}{h} && (31) \\ &= \lim_{h \to 0} \left(\frac{16}{\sqrt{2}} + \frac{9h}{2} + \frac{h^2}{2\sqrt{2}}\right) && (32) \\ &= \frac{16}{\sqrt{2}} && (33) \\ &= 8\sqrt{2}. && (34) \end{aligned}$$

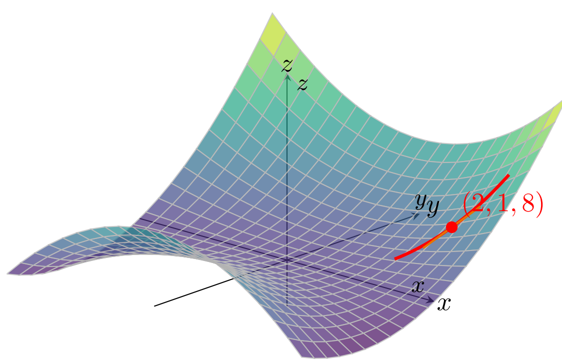


Figure 3: The surface $z = x^2 y + 4y^2$ together with the red curve obtained by moving from $(2, 1)$ in the direction $u = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$. The orange line is the tangent line, whose slope is the directional derivative $D_u f(2, 1) = 8\sqrt{2}$.

Example 1.5

Let
$$f(x, y, z, w) = x^2 y + z\cos(w). \tag{35}$$
We compute the directional derivative of f at the point $(1, 2, 0, 0)$ in the direction of the vector
$$v = (1, 1, 1, 1).$$
First, we normalize v. Since
$$\|v\| = \sqrt{1^2 + 1^2 + 1^2 + 1^2} = 2, \tag{36}$$
the unit direction vector is
$$u = \frac{v}{\|v\|} = \left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right). \tag{37}$$
By the limit definition of the directional derivative,
$$\begin{aligned} D_u f(1, 2, 0, 0) &= \lim_{h \to 0} \frac{f\big((1, 2, 0, 0) + hu\big) - f(1, 2, 0, 0)}{h} && (38) \\ &= \lim_{h \to 0} \frac{f\left(1 + \frac{h}{2}, 2 + \frac{h}{2}, \frac{h}{2}, \frac{h}{2}\right) - f(1, 2, 0, 0)}{h}. && (39) \end{aligned}$$
Now,
$$f(1, 2, 0, 0) = 1^2(2) + 0\cos(0) = 2. \tag{40}$$
Also,
$$\begin{aligned} f\left(1 + \frac{h}{2}, 2 + \frac{h}{2}, \frac{h}{2}, \frac{h}{2}\right) &= \left(1 + \frac{h}{2}\right)^2 \left(2 + \frac{h}{2}\right) + \frac{h}{2}\cos\left(\frac{h}{2}\right) && (41) \\ &= \left(1 + h + \frac{h^2}{4}\right)\left(2 + \frac{h}{2}\right) + \frac{h}{2}\cos\left(\frac{h}{2}\right) && (42) \\ &= 2 + \frac{5h}{2} + h^2 + \frac{h^3}{8} + \frac{h}{2}\cos\left(\frac{h}{2}\right). && (43) \end{aligned}$$
Therefore,
$$\begin{aligned} D_u f(1, 2, 0, 0) &= \lim_{h \to 0} \frac{2 + \frac{5h}{2} + h^2 + \frac{h^3}{8} + \frac{h}{2}\cos\left(\frac{h}{2}\right) - 2}{h} && (44) \\ &= \lim_{h \to 0} \left(\frac{5}{2} + h + \frac{h^2}{8} + \frac{1}{2}\cos\left(\frac{h}{2}\right)\right) && (45) \\ &= \frac{5}{2} + \frac{1}{2} && (46) \\ &= 3. && (47) \end{aligned}$$

But be careful! The existence of partials does not imply the existence of all directional derivatives!

Example 1.6 (Existence of Partials $\not\Longrightarrow$ Existence of Directional Derivatives)

Consider the function
$$f(x_1, x_2) = \begin{cases} \frac{x_1 x_2}{x_1^2 + x_2^2} & \text{if } (x_1, x_2) \neq (0, 0) \\ 0 & \text{if } (x_1, x_2) = (0, 0) \end{cases}$$

The partial derivatives exist everywhere. Away from the origin we can simply compute

$$\frac{\partial f}{\partial x_1} = \frac{x_2(x_1^2 + x_2^2) - x_1 x_2 \cdot 2x_1}{(x_1^2 + x_2^2)^2} = \frac{-x_1^2 x_2 + x_2^3}{(x_1^2 + x_2^2)^2}$$
$$\frac{\partial f}{\partial x_2} = \frac{x_1(x_1^2 + x_2^2) - x_1 x_2 \cdot 2x_2}{(x_1^2 + x_2^2)^2} = \frac{x_1^3 - x_1 x_2^2}{(x_1^2 + x_2^2)^2}$$

As for the partials at the origin, we must compute using the limit rule.

$$\partial_{x_1} f(0) = \lim_{h \to 0} \frac{f(0 + he_1) - f(0)}{h} = \lim_{h \to 0} \frac{f(h,0)}{h} = 0$$
$$\partial_{x_2} f(0) = \lim_{h \to 0} \frac{f(0 + he_2) - f(0)}{h} = \lim_{h \to 0} \frac{f(0,h)}{h} = 0$$

However, the directional derivative taken in direction $v = (1,1)$ gives

$$\begin{aligned} \nabla_{(1,1)} f(0,0) &= \lim_{h \to 0} \frac{f(0 + h(1,1)) - f(0)}{h} \\ &= \lim_{h \to 0} \frac{f(h,h)}{h} \\ &= \lim_{h \to 0} \frac{1}{2h} \end{aligned}$$

which does not have a limit as $h \to 0$. To visualize this, let's look at the values of f along various lines in $\mathbb{R}^2$.

1. $f = 0$ at the line $x_1 = 0$ and $x_2 = 0$, which is why the partials are 0.
2. $f = \frac{1}{2}$ at the line where $x_1 = x_2$, except for the point $(0,0)$, where $f = 0$, which is why the limit in the direction doesn't exist.

Definition 1.3 (Path Derivative)

Let $f : E \subset \mathbb{R}^n \to \mathbb{R}^m$, $a \in E$, and $p : \mathbb{R} \to \mathbb{R}^n$ be a path function where $p(t_0) = a$. Then, the **path derivative** of f at a with respect to path p is defined

$$D_p f(a) \coloneqq \lim_{t \to t_0} \frac{f(p(t)) - f(p(t_0))}{h} \tag{48}$$

Example 1.7

Let

$$f(x,y) = x^2 + y, \tag{49}$$

and let

$$p(t) = (1 + t, 1 + t^2). \tag{50}$$

Since

$$p(0) = (1,1), \tag{51}$$

we compute the path derivative of f at $a = (1,1)$ with respect to the path p.
By the definition of the path derivative,

$$D_p f(1,1) = \lim_{t \to 0} \frac{f(p(t)) - f(p(0))}{t - 0} \tag{52}$$
$$= \lim_{t \to 0} \frac{f(1 + t, 1 + t^2) - f(1,1)}{t}. \tag{53}$$

Now,

$$f(1,1) = 1^2 + 1 = 2, \tag{54}$$

and

$$\begin{aligned} f(1+t, 1+t^2) &= (1+t)^2 + (1+t^2) && (55)\\ &= 1 + 2t + t^2 + 1 + t^2 && (56)\\ &= 2 + 2t + 2t^2. && (57) \end{aligned}$$

Therefore,

$$\begin{aligned} D_p f(1,1) &= \lim_{t\to 0} \frac{2 + 2t + 2t^2 - 2}{t} && (58)\\ &= \lim_{t\to 0} \frac{2t + 2t^2}{t} && (59)\\ &= \lim_{t\to 0} (2 + 2t) && (60)\\ &= 2. && (61) \end{aligned}$$

Example 1.8

Let

$$f(x,y) = (xy, x^2 - y), \tag{62}$$

and let

$$p(t) = (\cos t, \sin t). \tag{63}$$

Since

$$p(0) = (1,0), \tag{64}$$

we compute the path derivative of f at $a = (1,0)$ with respect to the path p.
By the definition of the path derivative,

$$\begin{aligned} D_p f(1,0) &= \lim_{t\to 0} \frac{f(p(t)) - f(p(0))}{t - 0} && (65)\\ &= \lim_{t\to 0} \frac{f(\cos t, \sin t) - f(1,0)}{t}. && (66) \end{aligned}$$

Now,

$$f(1,0) = (0,1), \tag{67}$$

and

$$f(\cos t, \sin t) = \big(\cos t \sin t, \cos^2 t - \sin t\big). \tag{68}$$

Therefore,

$$\begin{aligned} D_p f(1,0) &= \lim_{t\to 0} \frac{\big(\cos t \sin t, \cos^2 t - \sin t\big) - (0,1)}{t} && (69)\\ &= \lim_{t\to 0} \left(\frac{\cos t \sin t}{t}, \frac{\cos^2 t - \sin t - 1}{t}\right). && (70) \end{aligned}$$

We compute each component separately. For the first component,

$$\begin{aligned} \lim_{t\to 0} \frac{\cos t \sin t}{t} &= \lim_{t\to 0} \cos t \cdot \frac{\sin t}{t} && (71)\\ &= 1. && (72) \end{aligned}$$

For the second component,

$$\lim_{t\to 0} \frac{\cos^2 t - \sin t - 1}{t} = \lim_{t\to 0} \left(\frac{\cos^2 t - 1}{t} - \frac{\sin t}{t} \right) \tag{73}$$
$$= \lim_{t\to 0} \left(\frac{-\sin^2 t}{t} - \frac{\sin t}{t} \right) \tag{74}$$
$$= 0 - 1 \tag{75}$$
$$= -1. \tag{76}$$

Hence,

$$D_p f(1,0) = (1,-1). \tag{77}$$

1.2 Total Derivatives

Definition 1.4 (Total/Frechet Derivative)

Let $f : E \subset \mathbb{R}^n \to \mathbb{R}^m$ and $a \in E$. f is **differentiable** at a if it satisfies either of the equivalent conditions.

1. *Linear Approximation.* There exists a linear map $Df_a : \mathbb{R}^n \longrightarrow \mathbb{R}^m$[a] such that

$$\lim_{h\to 0} \frac{\|f(a+h) - f(a) - Df_a h\|}{\|h\|} = 0 \tag{78}$$

2. *Fundamental Increment Lemma.* There exists a linear map $Df_a : \mathbb{R}^n \to \mathbb{R}^m$ and an infinitesimal[b] function $\Phi : \mathbb{R}^n \to \mathbb{R}^m$ defined for sufficiently small nonzero $h \in \mathbb{R}^n$ such that

$$f(a+h) = f(a) + Df_a h + \Phi(h)\|h\|, \qquad \lim_{h\to 0} \Phi(h) = 0 \tag{79}$$

If such a map Df_a exists, then it is unique, and we call this the **total derivative**—or **Frechet derivative**—of f at a.

[a] Note that when $m = 1$, then Df_a is a linear functional, i.e. a dual vector.
[b] As in $\Phi(h) \to 0$ as $h \to 0$.

Proof. We'll first prove that the two are equivalent.

1. $(\to)$. Given that linear approximation holds, we can construct the map

$$\Phi(h) := \frac{f(a+h) - f(a) - Df_a h}{\|h\|} \tag{80}$$

Therefore, by our assumption and absolute homogeneity of the norm, we have

$$0 = \lim_{h\to 0} \frac{\|f(a+h) - f(a) - Df_a h\|}{\|h\|} \tag{81}$$
$$= \lim_{h\to 0} \left\| \frac{f(a+h) - f(a) - Df_a h}{\|h\|} \right\| \tag{82}$$
$$= \lim_{h\to 0} \|\Phi(h)\| \tag{83}$$

Since the norm $v \mapsto \|v\|$ is continuous, this implies that

$$\lim_{h\to 0} \Phi(h) = 0 \tag{84}$$

2. ($\leftarrow$). Given that the fundamental increment lemma holds, we can see that for nonzero h,

$$\Phi(h) = \frac{f(a+h) - f(a) - Df_a h}{\|h\|} \tag{85}$$

Therefore, by taking the limit of both sides, we must have

$$0 = \lim_{h \to 0} \Phi(h) = \lim_{h \to 0} \frac{f(a+h) - f(a) - Df_a h}{\|h\|} \tag{86}$$

Therefore, by continuity of the norm and absolute homogeneity,

$$\begin{aligned} 0 = \|0\| &= \left\| \lim_{h \to 0} \frac{f(a+h) - f(a) - Df_a h}{\|h\|} \right\| && (87) \\ &= \lim_{h \to 0} \left\| \frac{f(a+h) - f(a) - Df_a h}{\|h\|} \right\| && (88) \\ &= \lim_{h \to 0} \frac{\|f(a+h) - f(a) - Df_a h\|}{\|h\|} && (89) \end{aligned}$$

and we are done.

Finally, to prove uniqueness of the derivative, it seems more convenient to use the fundamental increment lemma. Say that there are two derivatives, A_1 and A_2, each with their own infinitesimal functions Φ_1, Φ_2 defined in the intersection of the open neigborhoods of a, which is also an open neighborhood. Then,

$$\begin{aligned} f(a+h) &= f(a) + A_1 h + \Phi_1(h)\|h\| && (90) \\ f(a+h) &= f(a) + A_2 h + \Phi_2(h)\|h\| && (91) \end{aligned}$$

So, by subtracting them and rearranging, we have

$$0 = (A_1 - A_2)h + (\Phi_1 - \Phi_2)(h) \tag{92}$$

So for $h \neq 0$, we have

$$(A_2 - A_1)\frac{h}{\|h\|} = (\Phi_1 - \Phi_2)(h) \tag{93}$$

The left hand side is a linear map of h and is in fact a constant c since $\frac{h}{\|h\|}$ is a unit vector. So by taking the limits of both sides, we get

$$c = \lim_{h \to 0}(A_2 - A_1)\frac{h}{\|h\|} = \lim_{h \to 0}(\Phi_1 - \Phi_2)(h) = 0 \tag{94}$$

Therefore, $c = 0 \implies A_2 - A_1 = 0$. So $A_1 = A_2$.

The reason we want E to be open is that we require that $x + h$ to also be in D for sufficiently small h, and we can guarantee this since an ϵ-ball around x is guaranteed to be in E. As in univariate analysis, we first establish the relation to continuity.

Theorem 1.2 (Differentiability Implies Continuity)

If f is differentiable at x, then it is continuous at x.

Proof. By assumption, we have

$$\lim_{h\to 0} \frac{\|f(x+h) - f(x) - Df_x h\|}{\|h\|} = 0 \implies \lim_{h\to 0} f(x+h) - f(x) - Df_x h = 0 \tag{95}$$

since the norm is a continuous mapping. But $\lim_{h\to 0} Df_x h = 0$ since Df_x is linear, and so

$$\lim_{h\to 0} f(x+h) - f(x) = 0 \tag{96}$$

Therefore, we have 4 separate notions of derivatives. A natural question to ask is whether the existence of one derivative implies the existence of another.

Theorem 1.3 (Differentiability $\implies$ Existence of All Directionals)

If $f : E \subset \mathbb{R}^n \to \mathbb{R}^m$ is differentiable at $a \in E$, then all of its directional derivatives exist. Furthermore, given direction vector $u \in \mathbb{R}^n$,

$$D_u f_a = Df_a(u) \tag{97}$$

That is, we can compute the directional derivative by plugging in the direction vector into the linear map Df_a.

Proof. Since f is differentiable at a, the limit

$$\lim_{h\to 0} \frac{\|f(a+h) - f(a) - Df_a(h)\|}{\|h\|} = 0 \tag{98}$$

for all paths through 0. Therefore, let us choose the path

$$\lim_{h\to 0} \frac{\|f(a+hu) - f(a) - Df_a(hu)\|}{\|hu\|} = \lim_{h\to 0} \left\| \frac{f(a+hu) - f(a) - Df_a(hu)}{h} \right\| = 0 \in \mathbb{R} \tag{99}$$

Since the norm is a continuous mapping, this means that

$$0 = \lim_{h\to 0} \frac{f(a+hu) - f(a) - Df_a(hu)}{h} \tag{100}$$

$$= \lim_{h\to 0} \frac{f(a+hu) - f(a) - hDf_a(u)}{h} \tag{101}$$

$$= \lim_{h\to 0} \frac{f(a+hu) - f(a)}{h} - Df_a(u) \tag{102}$$

Therefore,

$$Df_a(u) = \lim_{h\to 0} \frac{f(a+hu) - f(a)}{h} = D_u f_a \tag{103}$$

Therefore, we know that differentiability (i.e. existence of the total derivative) is the strongest. If this is the case, then the partial and directional derivatives exist, and we also know how the derivative acts on each basis vector. Therefore, we can immediately know that

Lemma 1.4 (Action of Derivative on Basis Vectors)

Let $(e_i)_i$ and (e'_j) be the standard basis vectors of $\mathbb{R}^n$ and $\mathbb{R}^m$, respectively. If $f : \mathbb{R}^n \to \mathbb{R}^m$ is differentiable at a, then the partial derivatives $\frac{\partial f_j}{\partial x_i}$ exist, and

$$Df_a(e_i) = \sum_{j=1}^m \bigg(\frac{\partial f_j}{\partial x_i}(a) \bigg) e_j' \tag{104}$$

Proof. The partial derivatives existing is an immediate consequence of the previous theorem. As for the identity, we can see that

$$\begin{aligned}
Df_a(e_i) = D_{e_i} f_a = D_{e_i} \bigg(\sum_{j=1}^m f_j e_j' \bigg)_a & \qquad (105) \\
= \sum_{j=1}^m D_{e_i} (f_j e_j')_a & \qquad (106) \\
= \sum_{j=1}^m D_{e_i} (f_j)_a e_j' & \qquad (107) \\
= \sum_{j=1}^m \frac{\partial f_j}{\partial x_i}(a)\, e_j' & \qquad (108)
\end{aligned}$$

Since our vector spaces come with a basis, with this theorem, we can realize Df_x in a matrix form.

Definition 1.5 (Jacobian)

Given $f : E \subset \mathbb{R}^n \to \mathbb{R}^m$ differentiable at x, the **Jacobian** of f at x is the matrix realization of the total derivative, with entries being precisely the partial derivatives.

$$Jf_a = \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \dots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \dots & \frac{\partial f_m}{\partial x_n} \end{pmatrix} \tag{109}$$

Therefore, we can see that the derivative maps the ith basis vector $e_i \in \mathbb{R}^n$ to the ith column of our Jacobian matrix.

Analogous to the univariate case, a nice way to visualize the derivative is by looking at the tangent *plane*. In general, we have an *affine hyperplane*. Define this here. Unfortunately, the only types of functions that we can meaningful visualize are those mapping $f : \mathbb{R}^2 \to \mathbb{R}$.

Definition 1.6 (Tangent Hyperplane)

If $f : E \subset \mathbb{R}^n \to \mathbb{R}$ is differentiable at $a \in \mathbb{R}^n$, then the affine hyperplane tangent to the graph of f at $(a, f(a))$ is the set

$$P = \{(x, f(x) \colon z = f(a) + Df_a(x - a)\}. \tag{110}$$

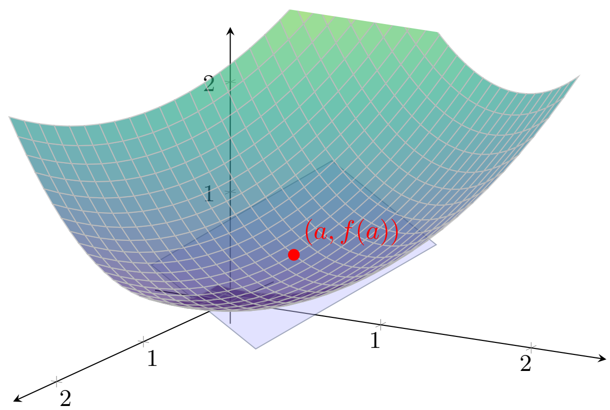


Figure 4: The tangent plane to the surface $z = f(x, y)$ at the point $(a, f(a))$. Here $f(x, y) = \frac{x^2+y^2}{2}$ and $a = (1, 1)$, so the tangent plane is $z = x + y - 1$.

So, to prove that a function $f : \mathbb{R}^n \longrightarrow \mathbb{R}$ is differentiable at a, and if so, what its total derivative is, there are essentially two steps.

1. We find a candidate for M by evaluating the partials.
$$M = \begin{pmatrix} \partial_{x_1} f(a) & \partial_{x_2} f(a) & \ldots & \partial_{x_n} f(a) \end{pmatrix} \tag{111}$$

2. We check to see if the limit is true.
$$\lim_{h \to 0} \frac{f(a + h) - f(a) - Mh}{||h||} = 0 \tag{112}$$
If it is, then $Df_a = M$, and the "tangent plane" of f at a is defined by the equation $y = f(a) + Mh$.

Example 1.9 (Computing Total Derivative)

The function $f(x_1, x_2) = x_1^2 + x_2^2$ is differentiable at $(1, 1)$. We let $M = \big(\partial_{x_1} f(1, 1), \partial_{x_2} f(1, 1)\big) = \big(2, 2\big)$ and see that

$$\lim_{h \to 0} \frac{f(a + h) - f(a) - Mh}{||h||} = \lim_{h \to 0} \frac{f(1 + h_1, 1 + h_2) - f(1, 1) - 2h_1 - 2h_2}{\sqrt{h_1^2 + h_2^2}} \tag{113}$$
$$= \lim_{h \to 0} \frac{(1 + h_1)^2 + (1 + h_2)^2 - 2 - 2h_1 - 2h_2}{\sqrt{h_1^2 + h_2^2}} \tag{114}$$
$$= \lim_{h \to 0} \sqrt{h_1^2 + h_2^2} = 0 \tag{115}$$

So, $Df(1, 1) = (2, 2)$.

Example 1.10 (Computing Tangent Plane)

Let us find the equation of the tangent plane to $f(x, y) = \ln(2x + y)$ at $(-1, 3)$. Our total derivative, if it exists, is the covector of partials

$$Df = \begin{pmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} \end{pmatrix} = \begin{pmatrix} \frac{2}{2x+y} & \frac{1}{2x+y} \end{pmatrix} \tag{116}$$

which is indeed continuous at a neighborhood of $(-1, 3)$ (in fact, in every neighborhood not containing 0). By continuity of partials, f is differentiable at $(-1, 3)$, with $Df_{(-1,3)} = (2, 1)$. The equation of the plane is then

$$z = f(-1,3) + Df_{(-1,3)} \begin{pmatrix} x+1 \\ y-3 \end{pmatrix} = 2(x+1) + 1(y-3) \implies z = 2x + y - 1 \tag{117}$$

However, the converse is not true for either statements. You cannot just evaluate all the partial derivatives and assume that the total derivative exists!

Example 1.11 (Partial Derivatives Exist, but Not Differentiable)

Consider the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined by:

$$f(x,y) = \begin{cases} \frac{xy}{x^2+y^2} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases} \tag{118}$$

At the origin $(0,0)$, the partial derivatives are:

$$\frac{\partial f}{\partial x}(0,0) = \lim_{h \to 0} \frac{f(h,0) - f(0,0)}{h} = \lim_{h \to 0} \frac{0-0}{h} = 0. \tag{119}$$

Similarly, $\frac{\partial f}{\partial y}(0,0) = 0$. Both partial derivatives exist. However, the total derivative **cannot** exist because the function is not continuous at $(0,0)$. As we saw in the limits section, approaching along $y = x$ gives a limit of $1/2$, while the function value is 0. A fundamental property of differentiability is that it implies continuity; therefore, f is not differentiable at the origin.

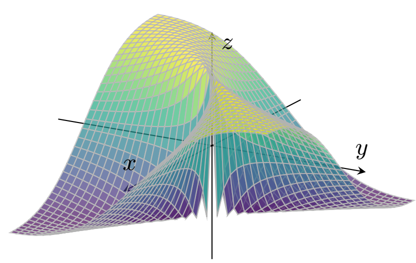


Figure 5: A function where partial derivatives exist at the origin, but the surface is "torn" there.

Example 1.12 (All Directional Derivatives Exist, but Not Differentiable)

Consider $f(x,y) = \begin{cases} \frac{x^2 y}{x^4+y^2} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$. For any direction $u = (u_1, u_2)$, the directional derivative at the origin exists and is given by $D_u f(0,0) = \frac{u_1^2}{u_2}$ (if $u_2 \neq 0$) or 0 (if $u_2 = 0$).

Despite having directional derivatives in every single direction, the total derivative does not exist. First, f is not continuous at the origin (approaching along $y = x^2$ gives $1/2 \neq 0$). Second, even if it were continuous, the total derivative requires that the linear map $Df(a)$ approximates the function well in *every* manner of approach, not just along straight lines. Here, the "peak" of the function follows the parabola $y = x^2$, which escapes the linear approximation provided by the directional derivatives.

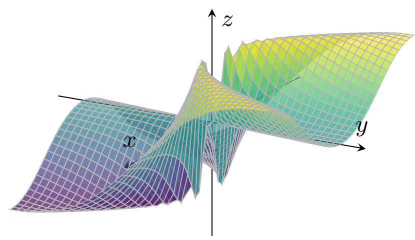


Figure 6: A function where all directional derivatives exist at the origin, yet the surface is non-differentiable.

1.3 Rules of Differentiation

Just like single variable calculus, the total derivative behaves in predictable ways: it is linear, product/quotient rules, and the chain rule.

Lemma 1.5 (Linearity of Total Derivatives)

Let $f, g : E \subset \mathbb{R}^n \longrightarrow \mathbb{R}^m$ be differentiable at $a \in E$. Then, the total derivative at a is linear w.r.t. the input functions.

1. *Addition.* $D(f+g)_a = Df_a + Dg_a$
2. *Scalar Multiplication.* $D(cf)_a = cDf_a$

Proof. Let f, g be differentiable at a and $c \in \mathbb{R}$.

1. *Addition.* There exists linear functions Df_a, Dg_a and infinitesimal functions Φ_f, Φ_g such that

$$f(a+h) = f(a) + Df_a(h) + \Phi_f(h)\|h\| \tag{120}$$
$$g(a+h) = g(a) + Dg_a(h) + \Phi_g(h)\|h\| \tag{121}$$

By adding the two, we get

$$(f+g)(a+h) = (f+g)(a) + (Df_a + Dg_a)(h) + (\Phi_f + \Phi_g)(h)\|h\| \tag{122}$$

where $Df_a + Dg_a$ is also linear and $\Phi_f + \Phi_g$ is also infinitesimal. So $f+g$ is differentiable with derivative $D(f+g)_a = Df_a + Dg_a$.

2. *Scalar Multiplication.* There exists a linear function Df_a and infintesimal Φ such that

$$f(a+h) = f(a) + Df_a(h) + \Phi(h)\|h\| \tag{123}$$

By multiplying both sides by c, we get

$$(cf)(a+h) = (cf)(a) + (cDf_a)(h) + (c\Phi)(h)\|h\| \tag{124}$$

where cDf_a is also linear and Φ also infinitesimal. So, cf is differentiable with derivative $D(cf)_a = cDf_a$.

Note that for the product and quotient rules, our scope is only for scalar valued functions since products and quotients aren't defined over codomains that are vector spaces.

Lemma 1.6 (Product Rule)

Let $f, g : E \subset \mathbb{R}^n \longrightarrow \mathbb{R}$ be differentiable at a. Then,

$$D(fg)_a = Df_a g(a) + f(a) Dg_a \tag{125}$$

Proof. By differentiability, we have

$$f(a+h) = f(a) + Df_a(h) + \Phi_f(h)\|h\| \tag{126}$$
$$g(a+h) = g(a) + Dg_a(h) + \Phi_g(h)\|h\| \tag{127}$$

By multiplying and expanding,[a] we get

$$\begin{aligned}(fg)(a+h) = f(a)g(a) + f(a)Dg_a(h) + g(a)Dg_a(h) && (128)\\ + Df_a(h)Dg_a(h) + f(a)\Phi_g(h)\|h\| + g(a)\Phi_g(h)\|h\| + Df_a(h)\Phi_g(h)\|h\| && (129)\\ + Dg_a(h)\Phi_f(h)\|h\| + \Phi_f(h)\|h\|\Phi_g(h)\|h\| && (130)\end{aligned}$$

The map $fDg_a + Df_a f$ is linear, so it suffices to prove that the rest of the terms can be turned into a form of $\Phi(h)\|h\|$ where Φ is infinitesimal. We can see that by linearity,

$$Df_a(h)Dg_a(h) = \frac{Df_a(h)Dg_a(h)\|h\|}{\|h\|} = Df_a(h)Dg_a(\frac{h}{\|h\|})\|h\| \tag{131}$$

where $\lim_{h\to 0} Df_a(h)Dg_a(\frac{h}{\|h\|}) = \big(\lim_{h\to 0} Df_a(h)\big)\big(\lim_{h\to 0} Dg_a(\frac{h}{\|h\|})\big) = 0 \cdot \big(\lim_{h\to 0} Dg_a(\frac{h}{\|h\|})\big) = 0$. By the limit rules, the rest of the terms are obvious.

[a] I wish there was a cleaner solution such as the proof for the univariate case, but I wasn't able to derive one.

Lemma 1.7 (Quotient Rule)

Let $f, g : E \subset \mathbb{R}^n \longrightarrow \mathbb{R}$ be differentiable at a with $g(a) \neq 0$. Then,

$$D\Big(\frac{f}{g}\Big)_a = \frac{Df_a g(a) - f(a)Dg_a}{g(a)^2} \tag{132}$$

Proof. Just like the univariate case, we may make an educated guess that $D(1/g)_a = -\frac{Dg_a}{g^2}$. Therefore,

$$\begin{aligned}\frac{(1/g)(a+h) - (1/g)(a) - \big(-\frac{Dg_a}{g^2}\big)}{\|h\|} &= \frac{\frac{g(a)^2 - g(a)g(a+h) + Dg_a(h)g(a+h)}{g(a+h)g(a)^2}}{\|h\|} && (133)\\ &= \frac{1}{g(a)^2}\frac{g(a)^2 - g(a)g(a+h) + Dg_a(h)g(a+h)}{\|h\|g(a+h)} && (134)\end{aligned}$$

We would like to turn this into a form that represents the limit definition of the derivative of g.

$$\begin{aligned}&= \frac{1}{g(a)^2}\frac{g(a)^2 - g(a+h)^2 + g(a+h)^2 - g(a)g(a+h) + Dg_a(h)g(a+h)}{\|h\|g(a+h)} && (135)\\ &= \frac{1}{g(a)^2}\Bigg(\underbrace{\frac{g(a)^2 - g(a+h)^2}{\|h\|g(a+h)}}_{\to 0} + \frac{g(a+h)\big[g(a+h) - g(a) - Dg_a(h)\big]}{\|h\|g(a+h)}\Bigg) && (136)\end{aligned}$$

The left term goes to 0 since g is differentiable and hence continuous. The right term can be simplified by canceling out the $g(a+h)$, which can be done since g is not 0 in a neighborhood of a. So, adding the limit back in, the whole term simplifies to

$$\begin{aligned}\lim_{h\to 0}\frac{(1/g)(a+h) - (1/g)(a) - \big(-\frac{Dg_a}{g^2}\big)}{\|h\|} &= \lim_{h\to 0}\frac{1}{g(a)^2}\frac{g(a+h) - g(a) - Dg_a(h)}{\|h\|} && (137)\\ &= \frac{1}{g(a)^2}\lim_{h\to 0}\frac{\|g(a+h) - g(a) - Dg_a(h)\|}{\|h\|} = 0 && (138)\end{aligned}$$

by the definition of g being differentiable at a.

Theorem 1.8 (Chain Rule)

Let $f : E \subset \mathbb{R}^n \longrightarrow \mathbb{R}^m$ be differentiable at $a \in E$ and $g : E' \subset \mathbb{R}^m \longrightarrow \mathbb{R}^p$ be differentiable at $g(a) \in E'$. Then, $g \circ f$ is differentiable at a, and

$$D(g \circ f)_a = Dg_{f(a)} \circ Df_a \tag{139}$$

Proof. Since f is differentiable at a and g at $f(a)$, we can write

$$f(a+h) = f(a) + Df_a(h) + \Phi_f(h)\|h\| \tag{140}$$
$$g(f(a)+t) = g(f(a)) + Dg_{f(a)}(t) + \Phi_g(t)\|t\| \tag{141}$$

Therefore, we wish to show that

$$\lim_{h \to 0} \frac{g(f(a+h)) - g(f(a)) - (Dg_{f(a)} \circ Df_a)(h)}{\|h\|} = 0 \tag{142}$$

Let's remove the limit and focus on the term. First substitute $f(a+h)$ and expand the term for $g(f(a)+t)$.

$$= \frac{g(f(a + \overbrace{Df_a(h) + \Phi_f(h)\|h\|}^{t})) - g(f(a)) - (Dg_{f(a)} \circ Df_a)(h)}{\|h\|} \tag{143}$$

$$= \frac{g(f(a)) + Dg_{f(a)}\big(Df_a(h) + \Phi_f(h)\|h\|\big) + \Phi\big(Df_a(h) + \Phi_f(h)\|h\|\big)\|h\| - g(f(a)) - Dg_{f(a)}(Df_a(h))}{\|h\|} \tag{144}$$

$$= \frac{(Dg_{f(a)} \circ Df_a)(h) + Dg_{f(a)}\big(\Phi_f(h)\|h\|\big) + \Phi_g\big(Df_a(h) + \Phi_f(h)\|h\|\big)\|h\| - (Dg_{f(a)} \circ Df_a)(h)}{\|h\|} \tag{145}$$

$$= Dg_{f(a)}\big(\Phi_f(h)\big) + \Phi_g\big(Df_a(h) + \Phi_f(h)\|h\|\big) \to 0 \tag{146}$$

where the final step follows from linearity of $Dg_{f(a)}$. As $h \to 0$, the limit of the left term is 0 since linear maps are continuous, and the right term goes to 0 since Φ_g is infinitesimal. We are done.

Therefore, given the composition of function $f \circ g$, we have two methods of finding the derivative matrix of $f \circ g$ at point x_0. First is to explicitly compute $f \circ g$ and find its $m \times p$ derivative matrix $D(f \circ g)$, and plug in a to get $D(f \circ g)_a$. The second way is to use the chain rule to find the individual total derivatives $Df_{g(a)}$ and Dg_a, and multiply them together.

1.4 Continuously Differentiable Functions

An even stronger condition beyond differentiability is continuous partials, and we often prove continuity of partials to prove differentiability.

Definition 1.7 (Continuously Differentiable)

A function $f : E \subset \mathbb{R}^n \to \mathbb{R}^m$ is **continuously differentiable** at $a \in E$ if it satisfies either of the equivalent definitions:

1. f is differentiable at a and the map $a \in \mathbb{R}^n \to Df_a \in \mathbb{R}^{m \times n}$ is continuous.

2. f is differentiable at a and all the partial derivatives $a \in \mathbb{R}^n \to \frac{\partial f_j}{\partial x_i}(a) \in \mathbb{R}$ is continuous.

Proof. We prove bidirectionally.

1. $(1 \to 2)$. Let $\phi_{ij} : \mathbb{R}^{m \times n} \to \mathbb{R}$ be the projection of the matrix onto the ijth element. This is continuous and we can see that
$$\frac{\partial f_j}{\partial x_i}(a) = \phi(Df_a) \tag{147}$$
so all partial derivatives are continous.
2. $(2 \to 1)$. Let $\epsilon > 0$. If all partial derivatives are continuous, then there exists a $\delta > 0$ such that
$$\|x - a\| < \delta \implies \left\| \frac{\partial f_j}{\partial x_i}(x) - \frac{\partial f_j}{\partial x_i}(a) \right\| < \frac{\epsilon}{mn} \implies \|Df_a\|_1 = \sum_{i,j} (Df_a)_{ij} < \epsilon \tag{148}$$
So $a \to Df_a$ is continuous with respect to the L_1 norm on $\mathbb{R}^{m \times n}$. But since all L_p norms are equivalent in finite-dimensional Euclidean space, it is continuous with respect to all norms.

Theorem 1.9 (Continuous Partials $\implies$ Differentiability)

Given a function $f : E \subset \mathbb{R}^n \longrightarrow \mathbb{R}$ and a point $a \in E$, if all the partials $\partial_{x_i} f$ exist and are continuous at a, then f is differentiable at a.

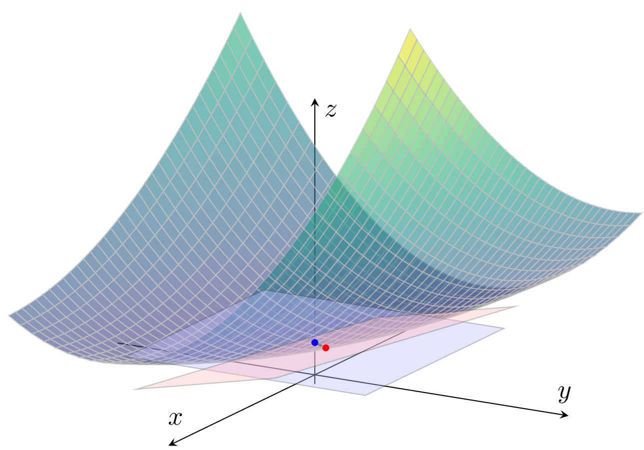


Figure 7: We can also visualize this theorem. Since the partials are continuous, then the tangent subspace, which is determined by the span of the tangent vectors determined by the partials, also changes continuously, and therefore the total derivative within a neighborhood of a exists.

Proof. The idea is to bound the perturbations in each component using the mean value theorem. Let us consider real valued $f : \mathbb{R}^n \to \mathbb{R}$ for now. Given $a, a + h \in \mathbb{R}^n$, let us construct a stepwise path from a to $a + h$. Let us denote $\phi_k(h) = (h_1, h_2, \ldots, h_k, 0, \ldots, 0)$. Then, we can construct the sequence
$$a = a + \phi_0(h), a + \phi_1(h), a + \phi_2(h), \ldots, a + \phi_k(h) = a + h \tag{149}$$
We take the matrix of partials
$$A = \begin{pmatrix} \frac{\partial f}{\partial x_1}(a) & \ldots & \frac{\partial f}{\partial x_n}(a) \end{pmatrix} \tag{150}$$

and wish to show that this is indeed the total derivative. We see that

$$f(a+h) - f(a) - Ah = f(a+h) - f(a) - \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}(a) h_i \tag{151}$$

$$= \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}(a) h_i + \big(f(a+\phi_i(h)) - f(a+\phi_{i-1}(h))\big) \tag{152}$$

Since $h_i \mapsto f(a+\phi_i(h)) - f(a+\phi_{i-1}(h))$ is a univariate differentiable function, by MVT, there exists a $c_i \in (a_i, a_i+h_i)$ s.t. $\frac{\partial f}{\partial x_i}(c_i) = \frac{f(a+\phi_i(h)) - f(a+\phi_{i-1}(h))}{h_i}$. Therefore, the above equals

$$= \sum_{i=1}^{n} \bigg(\frac{\partial f}{\partial x_i}(a) - \frac{\partial f}{\partial x_i}(c_i)\bigg) h_i \tag{153}$$

So,

$$\|f(a+h) - f(a) - Ah\| = \bigg\| \sum_{i=1}^{n} \bigg(\frac{\partial f}{\partial x_i}(a) - \frac{\partial f}{\partial x_i}(c_i)\bigg) h_i \bigg\| \tag{154}$$

Since $|h_i| \leq \|h\|$,

$$\frac{\|f(a+h) - f(a) - Ah\|}{\|h\|} = \bigg\| \sum_{i=1}^{n} \bigg(\frac{\partial f}{\partial x_i}(a) - \frac{\partial f}{\partial x_i}(c_i)\bigg) \frac{h_i}{\|h\|} \bigg\| \tag{155}$$

$$= \bigg\| \sum_{i=1}^{n} \bigg(\frac{\partial f}{\partial x_i}(a) - \frac{\partial f}{\partial x_i}(c_i)\bigg) \bigg\| \tag{156}$$

$$\leq \sum_{i=1}^{n} \bigg\| \frac{\partial f}{\partial x_i}(a) - \frac{\partial f}{\partial x_i}(c_i) \bigg\| \tag{157}$$

But from continuity of partials, we know that for $\epsilon > 0$, there exists a $\delta > 0$ s.t.

$$\|h\| < \delta \implies \bigg| \frac{\partial f}{\partial x_i}(a) - \frac{\partial f}{\partial x_i}(a - \phi_k(h)) \bigg| < \frac{\epsilon}{n} \tag{158}$$

$$\implies \bigg| \frac{\partial f}{\partial x_i}(a) - \frac{\partial f}{\partial x_i}(c_i) \bigg| < \frac{\epsilon}{n} \tag{159}$$

since $c_i \in (a, a + \phi_k(h))$. Therefore,

$$\frac{\|f(a+h) - f(a) - Ah\|}{\|h\|} = \sum_{i=1}^{n} \bigg| \frac{\partial f}{\partial x_i}(a) - \frac{\partial f}{\partial x_i}(c_i) \bigg| < \epsilon \tag{160}$$

While continuous partial derivatives guarantee differentiability, the converse is not true. A function can be differentiable even if its partial derivatives are not continuous. However, if they are not continuous, we cannot use the previous theorem to determine differentiability and must revert to the definition.

Example 1.13 (Non-Continuous Partials and Not Differentiable)

Recall $f(x,y) = \begin{cases} \frac{xy}{x^2+y^2} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$. The partial derivative $\frac{\partial f}{\partial x}$ for $(x,y) \neq (0,0)$ is:

$$\frac{\partial f}{\partial x} = \frac{y(x^2+y^2) - xy(2x)}{(x^2+y^2)^2} = \frac{y^3 - x^2 y}{(x^2+y^2)^2}.$$

At the origin, we found $\frac{\partial f}{\partial x}(0,0) = 0$. However, $\lim_{(x,y)\to(0,0)} \frac{y^3 - x^2 y}{(x^2+y^2)^2}$ does not exist (it depends on the path). Thus, the partial derivatives are not continuous at the origin. As shown before, the function is not differentiable because it is not continuous.

Example 1.14 (Non-Continuous Partials and Differentiable)

Consider $f(x,y) = \begin{cases} (x^2+y^2)\sin\left(\frac{1}{\sqrt{x^2+y^2}}\right) & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$. At the origin, the partial derivatives exist and are 0:

$$\frac{\partial f}{\partial x}(0,0) = \lim_{h\to 0} \frac{h^2 \sin(1/|h|) - 0}{h} = \lim_{h \to 0} h \sin(1/|h|) = 0.$$

Away from the origin, $\frac{\partial f}{\partial x}$ involves a $\cos(1/\sqrt{x^2+y^2}) \cdot \frac{1}{x^2+y^2}$ term which oscillates wildly as $(x,y) \to (0,0)$. Thus, the partials are not continuous.
However, f **is** differentiable at the origin. Using the definition with $Df(0,0) = 0$:

$$\lim_{h \to 0} \frac{|(h_1^2 + h_2^2)\sin(1/\|h\|) - 0 - 0|}{\|h\|} = \lim_{h\to 0} \|h\| \sin(1/\|h\|) = 0.$$

This satisfies the definition of the total derivative.

Theorem 1.10 (C^1 Space)

The set of all continuously differentiable functions $f : E \subset \mathbb{R}^n \to \mathbb{R}$, denoted $C^1(E)$, is a vector space.

Proof. The proof is trivial since Df is continuous. Therefore, sums and scalar multiples of are continuous as well.

Note that from now, whenever we talk about differentiating a function f, we will assume that it is C^1. This is overkill, since the set of all k-times differentiable functions is a subset of C^k, but it is conventional to work with C^k functions.

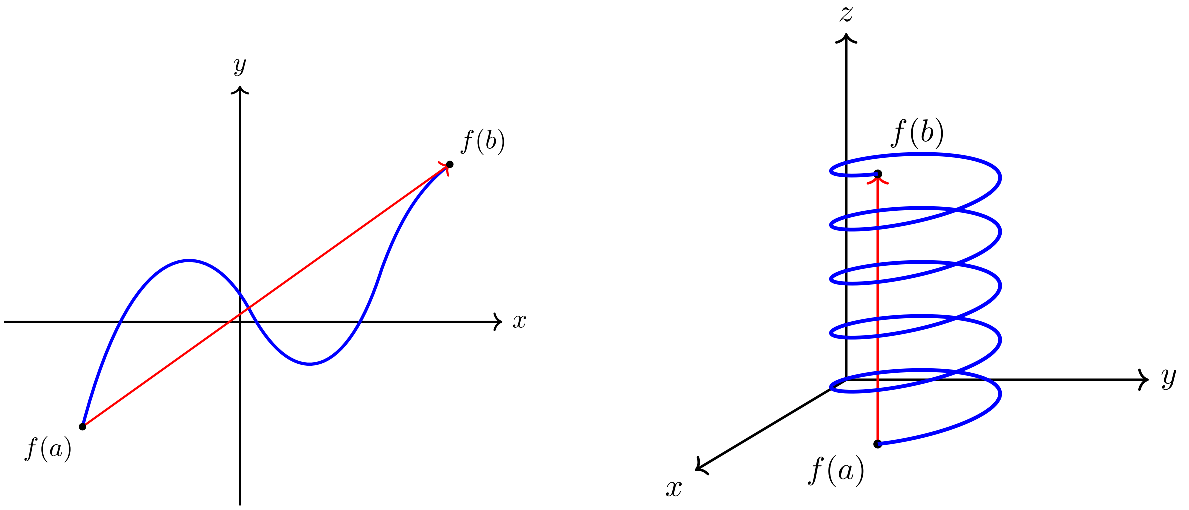
1.5 Mean Value Theorem

Lemma 1.11 (Mean Value Theorem for Paths)

If $f : E \subset \mathbb{R} \to \mathbb{R}^m$ is continuous on $[a,b]$ and differentiable on (a,b), then there exists $x \in (a,b)$ s.t.

$$\|f(b) - f(a)\| \leq (b-a)\|f'(x)\| \tag{161}$$

Proof. The general idea is that we want to convert this into a scalar-valued function in a clever way.



(a) (b)

Figure 8: If we sketch out a path in $\mathbb{R}^2$, we might speculate that at some point, the tangent line of the curve must be parallel to the secant vector $f(b) - f(a)$. Perhaps defining $\phi(x) = \langle f(b) - f(a), \frac{Df_x}{\|Df_x\|} \rangle$ and trying to maximize it using EVT or something might help. However, there is no guarantee that Df_x is continuous, and unfortunately, this assumption turns out not to be true as seen in the solenoid.

After some thought (see figure caption), we might think of looking at the inner product of the actual vectors on the curve with the secant vector, defining

$$\phi(x) = \langle f(b) - f(a), f(x) \langle \tag{162}$$

ϕ is differentiable on (a, b) and continuous on $[a, b]$, so by MVT, there exists some $x \in (a, b)$ such that

$$\phi(b) - \phi(a) = D\phi_x(b - a) = \langle f(b) - f(a), Df_x \rangle (b - a) \tag{163}$$

But $\phi(b) - \phi(a) = \langle f(b) - f(a), f(b) - f(a) \rangle = \|f(b) - f(a)\|^2$, so by Cauchy Schwarz, we get

$$\|f(b) - f(a)\|^2 = \langle f(b) - f(a), Df_x \rangle (b - a) \leq \|f(b) - f(a)\| \|Df_x\| (b - a) \tag{164}$$

Therefore, $\|f(b) - f(a)\| \leq \|Df_x\|(b - a)$.

There is a nice reason why equality doesn't hold in higher dimensions. For my first attempt at this proof, I tried to replicate my work for proving that C^1 implies differentiable by dividing f into component-wise paths. I defined

$$\phi_k(f) = \big(f_1(a), \ldots, f_k(a), f_{k+1}(b), \ldots, f_m(b) \big)^T \tag{165}$$

Therefore, I was hoping to bound $\|f(b) - f(a)\| \leq \sum_{k=1}^m \|\phi_k(f) - \phi_{k-1}(f)\| = \sum_{k=1}^m |f_k(b) - f_k(a)|$. But invoking the univariate MVT for the real-valued function f_k leads to finding multiple $x_k \in (a, b)$. This is a problem since we need to find one x.

Example 1.15 (MVT Inequality Does Not Hold on Circle)

Let

$$f : [0, 2\pi] \to \mathbb{R}^2, \quad f(x) = \big(\cos(x), \sin(x) \big) \tag{166}$$

Then, $f(2\pi) - f(0) = 0$, but

$$Df_x = \begin{pmatrix} -\sin(x) \\ \cos(x) \end{pmatrix} \tag{167}$$

is never 0 over $x \in [0, 2\pi]$.

Theorem 1.12 (Mean Value Theorem)

Let $f : E \subset \mathbb{R}^n \to \mathbb{R}^m$ be differentiable on convex open set E. If there exists a $M \in \mathbb{R}$ s.t. $\|Df_x\| \leq M$ for all $x \in E$, then

$$\|f(b) - f(a)\| \leq M\|b - a\| \tag{168}$$

for all $a, b \in E$.

Proof. Since E is convex, for any $a, b \in E$, we can consider

$$\gamma : [0,1] \to E, \quad \gamma(t) \coloneqq ta + (1-t)b \tag{169}$$

Then, $f \circ \gamma$ is a path function that is continuous on $[0,1]$ and differentiable on $(0,1)$. Therefore, there exists a $t_0 \in (0,1)$ (and hence, corresponding $x_0 = \gamma(t_0) \in E$) such that

$$\begin{aligned} \|(f\circ\gamma)(1) - (f\circ\gamma)(0)\| &\leq \|D(f\circ\gamma)_{t_0}\|\|1-0\| && (170) \\ &= \|Df_{x_0} D\gamma_{t_0}\| && (171) \\ &\leq \|Df_{x_0}(a-b)\| && (172) \\ &\leq \|Df_{x_0}\|\|a-b\| && (173) \\ &\leq M\|b-a\| && (174) \end{aligned}$$

where in the second step, we used the chain rule, in the third step, we computed the derivative $D\gamma = a - b$, and in the fourth step, we used the operator norm bound. The left hand side is really $\|f(b) - f(a)\|$, and we have our desired result.

The following corollary is pretty obvious.

Corollary 1.13 (Vanishing Derivative Implies Constant)

Let $f : E \subset \mathbb{R}^n \to \mathbb{R}^m$ be differentiable on convex open set E. If $\|Df_x\| = 0$ for all $x \in E$, then f is constant on E.

Proof. Set $M = 0$ in the proof above.

1.6 Inverse Function Theorem

Definition 1.8 (Contraction)

Given a metric space (X, d) and some $c < 1$, a **contraction** is a mapping $\phi : X \to X$ such that

$$d(\phi(x), \phi(y)) \leq c d(x, y) \tag{175}$$

Theorem 1.14 (Banach Fixed Point Theorem)

Given a complete metric space (X, d) and a contraction ϕ on X, there exists a unique point $x \in X$ such that $\phi(x) = x$.

Proof. We give a constructive proof on how to find such a point. Let us pick any $x_0 \in X$ and set $x_k = \phi^k(x_0)$ to be the output of the k-fold composition of ϕ. Then,

$$d(x_k, x_{k+1}) \leq c^k d(x_0, x_1) \tag{176}$$

So given any $m, n \in \mathbb{N}$, WLOG let $m < n$. Then,

$$\begin{aligned} d(x_m, x_n) &\leq \sum_{k=m}^{n-1} d(x_k, x_{k+1}) && (177) \\ &\leq \sum_{k=m}^{n-1} c^k d(x_0, x_1) && (178) \\ &= d(x_0, x_1) \sum_{k=m}^{n-1} c^k && (179) \\ &= d(x_0, x_1) \sum_{k=m}^{\infty} c^k && (180) \end{aligned}$$

Since the series converges from the geometric test, the tail sum $\sum_{k=m}^{\infty} c^k$ converges to 0 as $m \rightarrow +\infty$. So, $(x_n)_n$ is a Cauchy sequence and therefore converges to some $x \in X$.
Since ϕ is a contraction, ϕ is continuous. In fact it is uniformly continuous, since given $\epsilon > 0$, we can set $\delta = \frac{\epsilon}{c}$ and see that for all $x, y \in X$,

$$d(x, y) < \delta = \frac{\epsilon}{c} \implies d(\phi(x), \phi(y)) < c\frac{\epsilon}{c} = \epsilon \tag{181}$$

Therefore,

$$\phi(x) = \lim_{n \rightarrow +\infty} \phi(x_n) = \lim_{n \rightarrow +\infty} x_{n+1} = x \tag{182}$$

To prove uniqueness, assume that $x, y \in X$ are both fixed points. Then,

$$d(x, y) = d(\phi(x), \phi(y)) \leq c d(x, y) \implies d(x, y) = 0 \implies x = y \tag{183}$$

It is a bit strange that we just introduce this fixed point theorem all of a sudden when talking about invertibility, but this becomes useful in the proof for the inverse function theorem. When we have a point y and want to prove an inverse x exists such that $y = f(x)$, we can use the fixed point theorem to guarantee existence of such a point x.

Theorem 1.15 (Inverse Function Theorem)

Let $E \subset \mathbb{R}^n$ be an open neighborhood of a, and let $f : E \subset \mathbb{R}^n \rightarrow \mathbb{R}^n$ be C^1 over E. If Df_a is invertible, then

1. there exists open neighborhoods U of a and V of $b = f(a)$ such that

$$f|_U : U \longrightarrow V \tag{184}$$

is a bijection.

2. Moreover, the inverse map $(f|_U)^{-1} : V \to U$ is C^1 over V, and for every $y \in W$,

$$D(f^{-1})_y = (Df_x)^{-1}. \tag{185}$$

where $f(x) = y$.

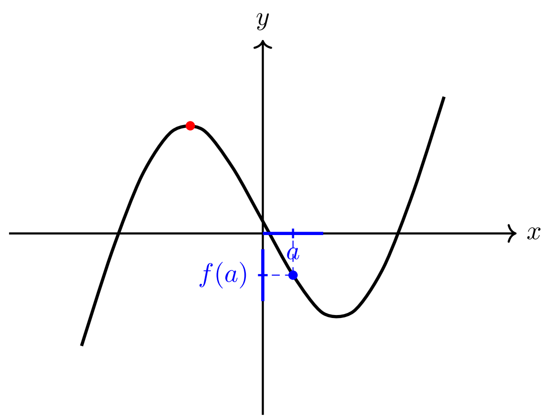


Figure 9: A cubic can fail to be locally invertible at a turning point, but behave regularly at nearby non-turning points.

Proof. We basically want to invert f, i.e. for each $y_0 \in \mathbb{R}^n$, we want to find a x_0 s.t. $f(x_0) = y_0$. How do we do this constructively? Since this may be difficult for an arbitrary function f, the next best thing to do is linearize $f(x)$ and find the inverse of the best linear approximation. By definition of differentiability at $a \in E$, there exists a linear map $Df_a : \mathbb{R}^n \to \mathbb{R}^n$ and infinitesimal function $\Phi : \mathbb{R}^n \to \mathbb{R}^n$ such that $f(x) = f(a) + Df_a(x-a) + \Phi(x-a)\|x-a\|$ within some neighborhood U_1 of a. Therefore, we can approximate $f(x)$ with a linear approximation $f_a(x)$, where the subscript is used to denote that it is centered at a.

$$y = f_a(x) \approx f(a) + Df_a(x-a) \tag{186}$$

After inverting this, we get the equation

$$x = g_a(y) = a + Df_a^{-1}(y - f(a)) \tag{187}$$

This function g basically takes the difference $y - f(x)$ in the codomain, inverts it to $Df_a^{-1}(y - f(x))$ as the difference in the domain, and finally adds it to x. This equation is a very natural approximation of the inverse, but the problem is that is that it is not an exact inverse. One quick fix is to set the center $a = x$ so that $f(x) = y$. Then, we have[a b]

$$f(x) = y \iff g_x(y) = x + Df_a^{-1}(y - f(x)) = x \tag{188}$$

Therefore, we have an equivalent notion of y being the image of x. Now rather than thinking about g_x as a function of y, we can think of it as a function of x. Define

$$\phi_y(x) \coloneqq g_x(y) \tag{189}$$

to be the "linear inverse of y centered at x." Then, note that

$$f(x_0) = y_0 \iff \phi_{y_0}(x_0) = g_{x_0}(y_0) = x_0 \iff x_0 \text{ is a fixed point of } \phi_{y_0} \tag{190}$$

The idea is to prove that ϕ_{y_0} is a contraction of some subset that is a complete metric space, and so (by the Banach fixed point theorem) the unique fixed point—call it $x_0 \in \mathbb{R}^n$—is the inverse. Let's do this step by step.

1. ϕ_0 *is a contraction.* $\phi_{y_0}(x) = x + Df_a^{-1}(y_0 - f(x))$, which implies that
$$D(\phi_{y_0})_x = I - Df_a^{-1} Df_x = Df_a^{-1}(Df_a - Df_x) \tag{191}$$
But since $x \to Df_x$ is continuous at a, there exists an open neighborhood U_2 containing a s.t.
$$x \in U_2 \implies \|Df_x - Df_a\| < \frac{1}{2\|Df_a^{-1}\|} \tag{192}$$
Therefore, let $U = U_1 \cap U_2$, so when $x \in U$,
$$\|D(\phi_{y_0})_x\| = \|Df_a^{-1}\| \|Df_a - Df_x\| \leq \|Df_a^{-1}\| \frac{1}{2\|Df_a^{-1}\|} = \frac{1}{2} \tag{193}$$
Therefore, by the MVT, for all $x_1, x_2 \in U$,
$$\|\phi_{y_0}(x_1) - \phi_{y_0}(x_2)\| \leq \frac{1}{2}\|x_1 - x_2\| \tag{194}$$
and so, ϕ_{y_0} is a contraction on U.
2. f *is injective over* U. Say that $x_1, x_2 \in U$ such that $f(x_1) = f(x_2) = y_0$. Then, from here, x_1 and x_2 must both be fixed points of U. Then,
$$\|x_1 - x_2\| = \|\phi_{y_0}(x_1) - \phi_{y_0}(x_2)\| \leq \frac{1}{2}\|x_1 - x_2\| \implies x_1 = x_2 \tag{195}$$
Therefore, there can at most be one such x_0 that maps to y_0. Hence f is injective and therefore $f : U \to f(U)$ is bijective, hence is invertible. Note that we did *not* use Banach fixed point theorem yet.
3. $V = f(U)$ *is open.* This is much trickier than it looks. First, pick $y_1 \in V$ and so $y_1 = f(x_1)$ for some $x_1 \in U$. We wish to show that there exists a δ-neighborhood around y_1, which establishes that V is open. The choose δ, let us choose $\epsilon > 0$ so small that $\overline{B_\epsilon(x_1)} \subset U$. Then, if $x \in B_\epsilon(x_1)$, then *for all* $y_0 \in V$, we have
$$\begin{aligned} \|\phi_{y_0}(x) - x_1\| &\leq \|\phi_{y_0}(x) - \phi_{y_0}(x_1)\| + \|\phi_{y_0}(x_1) - x_1\| && (196)\\ &\leq \frac{1}{2}\|x - x_1\| + \|Df_a^{-1}(y_0 - f(x_1))\| && (197)\\ &\leq \frac{\epsilon}{2} + \|Df_a^{-1}\|\|y_0 - f(x_1)\| && (198)\\ &= \frac{\epsilon}{2} + \|Df_a^{-1}\|\|y_0 - y_1\| && (199) \end{aligned}$$
[c] Therefore, this hints that we need to bound $\|y_0 - y_1\|$, so we can choose y_0 (and hence ϕ_{y_0}) s.t.
$$\|y_0 - y_1\| < \frac{\epsilon}{2\|Df_a^{-1}\|} = \delta \tag{200}$$
By plutting this into the inequality above, we can see that
$$x \in B(x_1, \epsilon), \|y_0 - y_1\| < \delta \implies \phi_{y_0}(x) \in B(x_1, \epsilon) \tag{201}$$
[d] Since ϕ_{y_0} is a contraction over U, it is also a contraction over $\overline{B_\epsilon(x_1)}$. Furthemore, since $\overline{B_\epsilon(x_1)}$ is closed in $\mathbb{R}^n$, it is complete, so by Banach fixed point theorem, ϕ_{y_0} has a unique fixed point, call it $x_0 \in \overline{B_\epsilon(x_1)}$. Since x_0 being a fixed point of $\phi_{y_0}(x)$ is equivalent to $y_0 = f(x_0)$, we can therefore say that the inverse $x_0 = f^{-1}(y_0) \in \overline{B(x_1, \epsilon)}$. Finally, we have
$$x_0 \in \overline{B_\epsilon(x_1)} \implies y_0 = f(x_0) \in f(B_\epsilon(x_1)) \subset f(U) = V \tag{202}$$

Great, so we have established that $f : U \to V$ is bijective and thus invertible. Now we wish to show that for $f : U \to V$, $D(f^{-1})_b = (Df_a)^{-1}$. Let $f(a+h) = f(a) + k$. Then , we can see that

$$\begin{aligned} f^{-1}(b+k) - f^{-1}(b) - Df_a^{-1}k &= (a+h) - a - Df_a^{-1}k && (203)\\ &= h - Df_a^{-1}k && (204)\\ &= -Df_a^{-1}(k - Df_a h) && (205)\\ &= -Df_a^{-1}\big(f(a+h) - f(a) - Df_a h\big) && (206) \end{aligned}$$

Therefore,

$$\|f^{-1}(b+k) - f^{-1}(b) - Df_a^{-1}k\| \leq \|Df_a^{-1}\|\|f(a+h) - f(a) - Df_a h\| \tag{207}$$

We hope to divide the left hand side by $\|k\|$, the right hand side by $\|h\|$, and take their limits to 0. So, we would like to find a bound between k and h. Recall that since ϕ_{y_0} is a contraction for any y_0,

$$\begin{aligned} \frac{1}{2}\|h\| &\geq \|\phi_{y_0}(x+h) - \phi_{y_0}(x)\| && (208)\\ &= \|h + Df_a^{-1}\big(f(x) - f(x+h)\big) && (209)\\ &= \|h - Df_a^{-1}k\| && (210)\\ &\geq \|h\| - \|Df_a^{-1}k\| && (211) \end{aligned}$$

Therefore, $\|h\| \leq 2\|Df_a^{-1}k\| \leq 2\|Df_a^{-1}\|\|k\|$, which implies that as $k \to 0$, then $h \to 0$. Therefore, by combining the two inequalities, we have

$$\frac{\|f^{-1}(b+k) - f^{-1}(b) - Df_a^{-1}k\|}{\|k\|} \leq 2\|Df_a^{-1}\|^2 \frac{\|f(a+h) - f(a) - Df_a h\|}{\|h\|} \tag{212}$$

Taking both sides as $k \to 0$ makes the right hand side vanish, since this implies that we are letting $h \to 0$. Therefore, the left hand limit vanishes, and by definition this means that $D(f^{-1})_b = Df_a^{-1}$. Finally, since inversion is a continuous operator, the chain of mappings $a \mapsto Df_a \mapsto Df_a^{-1}$ is continuous, and therefore the composition

$$a \mapsto D(f^{-1})_a = Df_a^{-1} \tag{213}$$

is also continuous, making f^{-1} continuously differentiable over V.

[a] where the reverse implication follows from Df_a being nonsingular.

[b] Note that the a in Df_a is *fixed* and does not change with respect to x.

[c] When I tried reproving it, I tried to bound $\|\varphi_{y_0}(x) - x_1\| = \|x - x_1 + Df_a^{-1}(y_0 - f(x))\| \leq \|x - x_1\| + \|Df_a^{-1}\|\|y_0 - f(x)\| < \epsilon + \|Df_a^{-1}\|\|y_0 - f(x)\|$. However, there still has a dependence on x, and we can't reduce it to just one choice of y_0.

[d] That is, as long as the center x is close enough to x_1 *and* y_0 is close enough to y_1, the approximate inverse of y_0 is close to x_1.

Example 1.16 (A Univariate Example)

Consider the function $f : \mathbb{R} \longrightarrow \mathbb{R}$ defined by

$$f(x) = x^2. \tag{214}$$

Its derivative is

$$Df_x = f'(x) = 2x. \tag{215}$$

Therefore, at any point $x_0 \neq 0$, we have

$$Df_{x_0} = 2x_0 \neq 0. \tag{216}$$

Hence Df_{x_0} is invertible as a linear map from $\mathbb{R}$ to $\mathbb{R}$.

By the inverse function theorem, for every $x_0 \neq 0$, there exists an open neighborhood V of x_0 such that

$$f|_V : V \longrightarrow f(V) \tag{217}$$

is invertible.
For example, at $x_0 = 1$, we can take a small interval around 1, such as

$$V = \left(\frac{1}{2}, \frac{3}{2}\right). \tag{218}$$

On this interval, $f(x) = x^2$ is injective, so it has a well-defined inverse

$$f^{-1}(y) = \sqrt{y}. \tag{219}$$

However, f is not invertible on all of $\mathbb{R}$, since

$$f(1) = 1 = f(-1). \tag{220}$$

Thus the inverse function theorem guarantees local invertibility near points where $f'(x_0) \neq 0$, but it does not guarantee global invertibility.

Example 1.17

Consider the vector-valued function $f : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ defined by

$$f(x_1, x_2) = \begin{pmatrix} e^{x_1} \cos(x_2) \\ e^{x_1} \sin(x_2) \end{pmatrix}$$

The total derivative/Jacobian matrix is

$$Jf(x_1, x_2) = \begin{pmatrix} e^{x_1} \cos(x_2) & -e^{x_1} \sin(x_2) \\ e^{x_1} \sin(x_2) & e^{x_1} \cos(x_2) \end{pmatrix} \implies \det Jf(x_1, x_2) = e^{2x_1} \cos^2(x_2) + e^{2x_1} \sin^2(x_2) = e^{2x_1}$$

Since the determinant e^{2x_1} is nonzero everywhere, Df_x is nonsingular. Thus, the theorem guarantees that for every point $x_0 \in \mathbb{R}^2$, there exists a neighborhood about x_0 over which f is invertible. However, this does not mean f is invertible over its entire domain: in this case f isn't even injective since it is periodic: e.g. the preimage of $(e, 0)$ contains $(1, 0)$ and $(1, 2\pi)$.

Example 1.18 (A Function from $\mathbb{R}^3$ to $\mathbb{R}^3$)

Consider the function $f : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ defined by

$$f(x_1, x_2, x_3) = \begin{pmatrix} e^{x_1} \cos(x_2) \\ e^{x_1} \sin(x_2) \\ x_3 \end{pmatrix}. \tag{221}$$

The total derivative/Jacobian matrix is

$$Jf(x_1, x_2, x_3) = \begin{pmatrix} e^{x_1} \cos(x_2) & -e^{x_1} \sin(x_2) & 0 \\ e^{x_1} \sin(x_2) & e^{x_1} \cos(x_2) & 0 \\ 0 & 0 & 1 \end{pmatrix}. \tag{222}$$

Therefore,

$$\det Jf(x_1, x_2, x_3) = e^{2x_1} \cos^2(x_2) + e^{2x_1} \sin^2(x_2). \tag{223}$$

Hence

$$\det Jf(x_1, x_2, x_3) = e^{2x_1}. \tag{224}$$

Since $e^{2x_1} \neq 0$ for all $x_1 \in \mathbb{R}$, the derivative Df_x is invertible at every point $x \in \mathbb{R}^3$.
Thus, by the inverse function theorem, for every point $x_0 \in \mathbb{R}^3$, there exists an open neighborhood V of x_0 such that

$$f|_V : V \longrightarrow f(V) \tag{225}$$

is invertible.
However, f is not invertible on all of $\mathbb{R}^3$, since it is periodic in the x_2-variable. For example,

$$f(1, 0, 5) = \begin{pmatrix} e \\ 0 \\ 5 \end{pmatrix} = f(1, 2\pi, 5). \tag{226}$$

Therefore, f is locally invertible everywhere, but not globally invertible.

1.7 Implicit Function Theorem

Remember that given an explicit representation of a set $y = f(x)$, we can easily find the implicit form as $F(x, y) = y - f(x) = 0$. What about the other way around? That is, given an implicit representation of some surface, what conditions must be met so that it can be represented as the graph of a function? The implicit function theorem is a tool that allows relations between points in $\mathbb{R}^n$ to be converted to functions of several real variables. That is, it states that for sufficiently "nice" points on a n-dimensional surface defined as $F(x, y) = 0$ (where $F : \mathbb{R}^{n+m} \longrightarrow \mathbb{R}^m$), we can locally pretend that this surface is a graph of a function $g : \mathbb{R}^n \longrightarrow \mathbb{R}^m$ whose graph $\big(x, g(x)\big)$ is precisely the set of all (x, y) s.t. $f(x, y) = 0$. When $m = 1$, it basically states that if an implicit surface suffices the vertical line test in a neighborhood, then it can be written as a function.

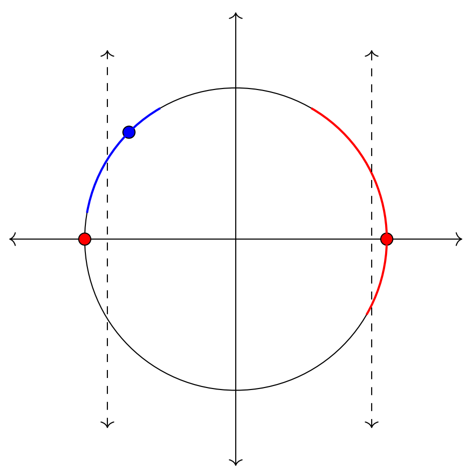


Figure 10: Let $f : \mathbb{R}^2 \longrightarrow \mathbb{R}$ be defined by $f(x, y) = x^2 + y^2 - 1$. The level set at $z = 0$ would be the set of points satisfying $x^2 + y^2 - 1 = 0$, i.e. the unit circle. The derivative of f with respect to y is 0 at the points $(-1, 0)$ and $(1, 0)$, meaning that in any neighborhood of these points, we cannot define a function of y with respect to x. This is true, indeed, since any such function would fail the vertical line test, which can be seen in the red neighborhood around $(1, 0)$. However, the blue neighborhood of the point $(-\sqrt{2}/2, \sqrt{2}/2)$ does indeed define a function of y with respect to x satisfying the vertical line test.

Theorem 1.16 (Implicit Function Theorem)

Let $F : E \subset \mathbb{R}^{n+m} \to \mathbb{R}^m$ be C^1 over an open set E. Write points in $\mathbb{R}^{n+m}$ as (x, y), where $x \in \mathbb{R}^n$ and $y \in \mathbb{R}^m$. Given $(a, b) \in E$, if

1. $F(a, b) = 0$,
2. $D_y F_{(a,b)} \in \mathbb{R}^{m \times m}$ is invertible,

then there exist open neighborhoods $U \ni a$ and $V \ni b$ where $U \times V \subset E$, and a unique C^1 function $f : U \longrightarrow V$ such that $f(a) = b$ and, for every $(x, y) \in U \times V$,

$$F(x, y) = 0 \iff y = f(x). \tag{227}$$

Moreover, for every $x \in U$,

$$Df_x = -\left(D_y F_{(x,f(x))}\right)^{-1} D_x F_{(x,f(x))}. \tag{228}$$

Proof. The idea is quite simple. We augment F to a locally invertible function G, then take the inverse of it, and finally project it back to the y coordinates. Let us define

$$G : E \subset \mathbb{R}^{n+m} \to \mathbb{R}^{n+m}, \quad G(x, y) = (x, F(x, y)) \tag{229}$$

Note that $G(a, b) = (a, F(a, b)) = (a, 0)$, and to evaluate the derivative of G, we can find its Jacobian in block form:

$$DG_{(a,b)} = \begin{pmatrix} I_n & 0 \\ D_x F_{(a,b)} & D_y F_{(a,b)} \end{pmatrix}. \tag{230}$$

This is a block lower triangular matrix, with both block diagonal elements invertible, hence $\det DG_{(a,b)} = \det(I_n) \det(D_y F_{(a,b)}) = \det(D_y F_{(a,b)}) \neq 0$, and so $DG_{(a,b)}$ is invertible. By the inverse function theorem, there exist open neighborhoods W of (a, b) and O of $(a, 0)$ such that $G|_W : W \to O$ is a bijection with a C^1 inverse $(G|_W)^{-1} : O \to W$.

1. Since W is an open neighborhood of (a, b), we may choose open neighborhoods U_0 of a and V of b such that
$$U_0 \times V \subset W. \tag{231}$$
2. Since $(G|_W)^{-1}(a, 0) = (a, b) \in U_0 \times V$ and $U_0 \times V$ is open, we can take the preimage of it with respect to the continuous inverse $(G|_W)^{-1}$ to see that
$$\left((G|_W)^{-1}\right)^{-1}(U_0 \times V) = G|_W(U_0 \times V) \tag{232}$$
is an open neighborhood of $G|_W(a, b) = (a, 0)$. Therefore, you can choose an open neighborhood $U \ni a$ s.t.
$$U \times \{0\} \subset G|_W(U_0 \times V) \tag{233}$$
Then for every $x \in U$, we see that $(G|_W)^{-1}(x, 0) \in U_0 \times V$. This will be useful for setting up our proper codomain for our function f later.

Now that we have our desired open set U, we define[a]

$$f : U \to V, \qquad f(x) \coloneqq \pi_y\big((G|_W)^{-1}(x, 0)\big) \tag{234}$$

Note that since $x \in U$, $G|_W^{-1}(x, 0) \in U_0 \times V$ and so its y-projection $f(x) \in V$. It is also clear that f is C^1 since $(G|_W)^{-1}$ is C^1. Note that $f(a) = b$ because $G(a, b) = (a, 0)$, and $f(x)$ is the unique vector such that $(G|_W)^{-1}(x, 0) = (x, f(x))$. Now we must show that $y = f(x) \iff F(x, y) = 0$.

1. $(\rightarrow)$. Let $y = f(x)$, then $y = \pi_y\big((G|_W)^{-1}(x, 0)\big)$. By definition of projection, we must have $(G|_W)^{-1}(x, 0) = (*, y)$, but $* = x$ since neither G nor the inverse affects the x element. So,
$$(G|_W)^{-1}(x, 0) = (x, y) \implies (x, 0) = G|_W(x, y) = (x, F(x, y)) \implies 0 = F(x, y) \tag{235}$$

2. ($\leftarrow$). Now let $F(x,y)=0$. Then, for $(x,y) \in U \times V$, we have

$$G|_W(x,y) = \big(x, F(x,y)\big) = (x,0) \tag{236}$$

Since $G|_W$ is invertible, we can apply $(G|_W)^{-1}$ to both sides and so $(x,y) = (G|_W)^{-1}(x,0)$. By applying the projection operator to both sides, we have

$$y = \pi_y(x,y) = \pi_y\big((G|_W)^{-1}(x,0)\big) = f(x) \tag{237}$$

This also proves uniqueness. Indeed, if another function $\widetilde{f} : U \to V$ satisfies $F(x, \widetilde{f}(x)) = 0$ for all $x \in U$, then by the equivalence above, $\widetilde{f}(x) = f(x)$ for all $x \in U$.
It remains to compute the derivative of f. Since $F(x, f(x)) = 0$ for every $x \in U$, we differentiate both sides with respect to x. By the chain rule,

$$D_x F_{(x,f(x))} + D_y F_{(x,f(x))} Df_x = 0 \implies D_y F_{(x,f(x))} Df_x = -D_x F_{(x,f(x))}. \tag{238}$$

Since $D_y F_{(a,b)}$ is invertible, the set of invertible maps in $\mathcal{L}(\mathbb{R}^m, \mathbb{R}^m)$ is an open set, and the map $(x,y) \mapsto D_y F_{(x,y)}$ is continuous, there exists a neighborhood around (a,b) such that if $(x, f(x))$ is close enough to (a,b), then $D_y F_{(x,f(x))}$ is invertible. Therefore, we can shrink U accordingly to be in this neighborhood. Thus, in this new U, V, we have for all $x \in U$

$$Df_x = -\big(D_y F_{(x,f(x))}\big)^{-1} D_x F_{(x,f(x))}. \tag{239}$$

[a] π_y denotes projection onto the $\mathbb{R}^m$ component.

Example 1.19 (Circle Example)

Let $n = c = 1$, and consider the function $F : \mathbb{R} \times \mathbb{R} \longrightarrow \mathbb{R}$ defined by

$$F(x_1, x_2) = x_1^2 + x_2^2 - 1. \tag{240}$$

The level set

$$F(x_1, x_2) = 0 \tag{241}$$

is the unit circle

$$x_1^2 + x_2^2 = 1. \tag{242}$$

We would like to determine at which points of the circle we can locally write x_2 as a function of x_1. The total derivative of F is

$$DF_{(x_1,x_2)} = \begin{pmatrix} \partial_{x_1} F & \partial_{x_2} F \end{pmatrix} = \begin{pmatrix} 2x_1 & 2x_2 \end{pmatrix}. \tag{243}$$

In this case, if we want to solve for x_2 as a function of x_1, then we look at the partial derivative with respect to x_2:

$$D_{x_2} F_{(x_1,x_2)} = 2x_2. \tag{244}$$

This is invertible as a 1×1 matrix precisely when

$$x_2 \neq 0. \tag{245}$$

Therefore, by the implicit function theorem, at every point of the circle with $x_2 \neq 0$, we can locally write the circle in the form

$$x_2 = g(x_1). \tag{246}$$

For example, consider the point

$$\left(\frac{4}{5}, \frac{3}{5}\right) \tag{247}$$

on the unit circle. Since $\frac{3}{5} \neq 0$, we can locally solve for x_2 as a function of x_1. Near this point, the appropriate branch is

$$x_2 = g(x_1) = \sqrt{1 - x_1^2}. \tag{248}$$

On the other hand, at the points $(1, 0)$ and $(-1, 0)$, we have

$$D_{x_2} F_{(x_1,x_2)} = 2x_2 = 0. \tag{249}$$

So the implicit function theorem does not allow us to solve for x_2 as a function of x_1 near these points. Indeed, near $x_1 = 1$ or $x_1 = -1$, the circle fails to be the graph of a function of x_1, since nearby x_1-values correspond to two possible x_2-values.
The derivative of the implicit function g is given by the theorem:

$$Dg = -\left(D_{x_2} F\right)^{-1} D_{x_1} F. \tag{250}$$

Therefore,

$$Dg = -(2x_2)^{-1}(2x_1) = -\frac{x_1}{x_2}. \tag{251}$$

Since $x_2 = g(x_1)$, this becomes

$$g'(x_1) = -\frac{x_1}{g(x_1)}. \tag{252}$$

Similarly, if we want to solve for x_1 as a function of x_2, then we look at the partial derivative with respect to x_1:

$$D_{x_1} F_{(x_1,x_2)} = 2x_1. \tag{253}$$

This is invertible precisely when

$$x_1 \neq 0. \tag{254}$$

Hence, at every point of the circle with $x_1 \neq 0$, we can locally write the circle in the form

$$x_1 = h(x_2). \tag{255}$$

The derivative of h is

$$Dh = -\left(D_{x_1} F\right)^{-1} D_{x_2} F. \tag{256}$$

Therefore,

$$Dh = -(2x_1)^{-1}(2x_2) = -\frac{x_2}{x_1}. \tag{257}$$

Since $x_1 = h(x_2)$, this becomes

$$h'(x_2) = -\frac{x_2}{h(x_2)}. \tag{258}$$

Example 1.20 (Sphere Example)

Let $n = 2$ and $c = 1$, and consider the function $F : \mathbb{R}^2 \times \mathbb{R} \longrightarrow \mathbb{R}$ defined by

$$F(x_1, x_2, y) = x_1^2 + x_2^2 + y^2 - 1. \tag{259}$$

The level set

$$F(x_1, x_2, y) = 0 \tag{260}$$

is the unit sphere

$$x_1^2 + x_2^2 + y^2 = 1. \tag{261}$$

We would like to determine at which points of the sphere we can locally write y as a function of (x_1, x_2).

The total derivative of F is

$$DF_{(x_1,x_2,y)} = \begin{pmatrix} \partial_{x_1} F & \partial_{x_2} F & \partial_y F \end{pmatrix} = \begin{pmatrix} 2x_1 & 2x_2 & 2y \end{pmatrix}. \tag{262}$$

If we want to solve for y as a function of (x_1, x_2), then we look at the partial derivative with respect to y:

$$D_y F_{(x_1,x_2,y)} = 2y. \tag{263}$$

This is invertible as a 1×1 matrix precisely when

$$y \neq 0. \tag{264}$$

Therefore, by the implicit function theorem, at every point of the sphere with $y \neq 0$, we can locally write the sphere in the form

$$y = g(x_1, x_2). \tag{265}$$

For example, consider the point

$$\left(\frac{1}{2}, \frac{1}{2}, \frac{1}{\sqrt{2}} \right) \tag{266}$$

on the unit sphere. Since $\frac{1}{\sqrt{2}} \neq 0$, we can locally solve for y as a function of (x_1, x_2). Near this point, the appropriate branch is

$$y = g(x_1, x_2) = \sqrt{1 - x_1^2 - x_2^2}. \tag{267}$$

On the other hand, at points on the equator where $y = 0$, the implicit function theorem does not allow us to solve for y as a function of (x_1, x_2). Geometrically, near such points, the sphere has both an upper and a lower branch over the same nearby (x_1, x_2)-values.
The derivative of the implicit function g is given by

$$Dg = -\left(D_y F\right)^{-1} D_x F. \tag{268}$$

In this case,

$$D_x F = \begin{pmatrix} 2x_1 & 2x_2 \end{pmatrix} \tag{269}$$

and

$$D_y F = 2y. \tag{270}$$

Therefore,

$$Dg = -(2y)^{-1} \begin{pmatrix} 2x_1 & 2x_2 \end{pmatrix} = \begin{pmatrix} -\frac{x_1}{y} & -\frac{x_2}{y} \end{pmatrix}. \tag{271}$$

Since $y = g(x_1, x_2)$, this becomes

$$Dg_{(x_1,x_2)} = \begin{pmatrix} -\frac{x_1}{g(x_1,x_2)} & -\frac{x_2}{g(x_1,x_2)} \end{pmatrix}. \tag{272}$$

Example 1.21 (A System of Two Equations)

Let $n = 2$ and $c = 2$, and consider the function $F : \mathbb{R}^2 \times \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ defined by

$$F(x_1, x_2, y_1, y_2) = \begin{pmatrix} y_1^2 + y_2 - x_1 \\ y_1 + e^{y_2} - x_2 \end{pmatrix}. \tag{273}$$

We would like to locally solve the system

$$F(x_1, x_2, y_1, y_2) = 0 \tag{274}$$

for (y_1, y_2) as a function of (x_1, x_2).

First observe that

$$F(1,2,1,0) = \begin{pmatrix} 1^2 + 0 - 1 \\ 1 + e^0 - 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}. \tag{275}$$

Thus the point

$$(x_1, x_2, y_1, y_2) = (1,2,1,0) \tag{276}$$

lies on the zero set of F.
The derivative of F with respect to the y-variables is

$$D_y F_{(x_1,x_2,y_1,y_2)} = \begin{pmatrix} \partial_{y_1} F_1 & \partial_{y_2} F_1 \\ \partial_{y_1} F_2 & \partial_{y_2} F_2 \end{pmatrix} = \begin{pmatrix} 2y_1 & 1 \\ 1 & e^{y_2} \end{pmatrix}. \tag{277}$$

At the point $(1,2,1,0)$, this becomes

$$D_y F_{(1,2,1,0)} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}. \tag{278}$$

Its determinant is

$$\det D_y F_{(1,2,1,0)} = 2 \cdot 1 - 1 \cdot 1 = 1. \tag{279}$$

Therefore $D_y F_{(1,2,1,0)}$ is invertible.
By the implicit function theorem, there exist open neighborhoods U of $(1,2)$ and V of $(1,0)$, and a unique C^1 function

$$h : U \longrightarrow V \tag{280}$$

such that

$$h(x_1,x_2) = \begin{pmatrix} h_1(x_1,x_2) \\ h_2(x_1,x_2) \end{pmatrix} \tag{281}$$

and

$$F(x_1, x_2, h_1(x_1,x_2), h_2(x_1,x_2)) = 0 \tag{282}$$

for all $(x_1,x_2) \in U$.
In other words, near $(x_1,x_2) = (1,2)$, the system

$$\begin{cases} y_1^2 + y_2 = x_1, \\ y_1 + e^{y_2} = x_2 \end{cases} \tag{283}$$

implicitly defines (y_1,y_2) as a function of (x_1,x_2).
To compute the derivative of h, we use

$$Dh_x = -\left(D_y F_{(x,h(x))}\right)^{-1} D_x F_{(x,h(x))}. \tag{284}$$

In this example,

$$D_x F_{(x_1,x_2,y_1,y_2)} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{285}$$

Therefore,

$$Dh_x = \left(D_y F_{(x,h(x))}\right)^{-1}. \tag{286}$$

At the point $(x_1,x_2) = (1,2)$, we have $h(1,2) = (1,0)$, so

$$Dh_{(1,2)} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}^{-1}. \tag{287}$$

Hence

$$Dh_{(1,2)} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}. \tag{288}$$

1.8 Extrema

Recall what a local extrema is from topology. We restate it here.

Definition 1.9 (Local Extrema)

Given a function $f : E \subset \mathbb{R}^n \longrightarrow \mathbb{R}$,

1. $a \in D$ is a **local minimum** if there exists a neighborhood U of a such that

$$f(x) \geq f(a) \text{ for every } x \in U \tag{289}$$

2. $a \in D$ is a **local maximum** if there exists a neighborhood U of x such that

$$f(x) \leq f(a) \text{ for every } x \in U \tag{290}$$

Theorem 1.17 (First Derivative Test)

Let $E \subset \mathbb{R}^n$ be open, with $f : E \to \mathbb{R}$ differentiable at $a \in E$. If a is a local extremum of f, then $Df_a = 0$.

Proof. Unlike the sequential proof we did for the univariate case, we prove the contrapositive. WLOG, let a be a maximum and $Df_a \neq 0$. Then, we basically have a nontrivial dual vector $Df_a \in (\mathbb{R}^n)^*$, and so there exists a $v \in \mathbb{R}^n$ s.t. $Df_a v > 0$. Now by definition of differentiability, we have $f(a+v) = f(a) + Df_a v + \Phi(\|v\|)\|v\|$, so let us make the norm of v so small such that

$$\frac{Df_a v}{\|v\|} > |\Phi(\|v\|)| \tag{291}$$

which is possible since $\frac{Df_a v}{\|v\|}$ is a constant while $\Phi(\|v\|) \to 0$ as $\|v\| \to 0$. Let us also shrink v so that $a + v \in E$ as well. Therefore, we can find a $v \in \mathbb{R}^n$ so that $Df_a v > |\Phi(\|v\|)|\|v\|$, and so

$$f(a+v) = f(a) + \underbrace{Df_a v + \Phi(\|v\|)\|v\|}_{>0} > f(a) \tag{292}$$

This can be done for an arbitrary neighborhood of a, contradicting the fact that a is a local maximum.

Note that even though the converse of this theorem is not true, we can use the contrapositive to determine that every point that has a nonzero derivative cannot be a extremum.

Example 1.22 (Function with an Infinite Number of Critical Points)

A function may also have an infinite amount of critical points (e.g. if they lie in a circle).

Unfortunately, determining whether a point a is a local extremum requires us to define an open neighborhood around a. This means that we can only determine local extrema within open sets in $\mathbb{R}^n$. Therefore, we must modify our procedure when looking for extrema on functions defined over closed bounded sets. We now describe a method of computing the global extrema. Let $f : E \subset \mathbb{R}^n \longrightarrow \mathbb{R}$ be a multivariable function defined on a closed and bounded set $E = E^\circ \cup \partial E$, where the E° is the interior and ∂U is the boundary of E. To find the global extrema on E, we find all local extrema of

1. f defined over the interior of E°, which is open, by locating all points where $Df = 0$, and
2. f defined over ∂E.

In the second point, we could reparamaterize ∂E cleverly by another function $\phi : \mathbb{R}^k \to \mathbb{R}^n$ (e.g. a path function for 1-dimensional manifolds ∂E) where the image coincides with ∂E, but this can be tedious. A more flexible method is to introduce a system of equality constraints of the form

$$g : \mathbb{R}^n \to \mathbb{R}^c, \quad g(x) = \begin{pmatrix} g_1(x) \\ \vdots \\ g_c(x) \end{pmatrix} = 0 \tag{293}$$

which really just represents a level set of g at 0, i.e. a set described by an implicit representation.[1]

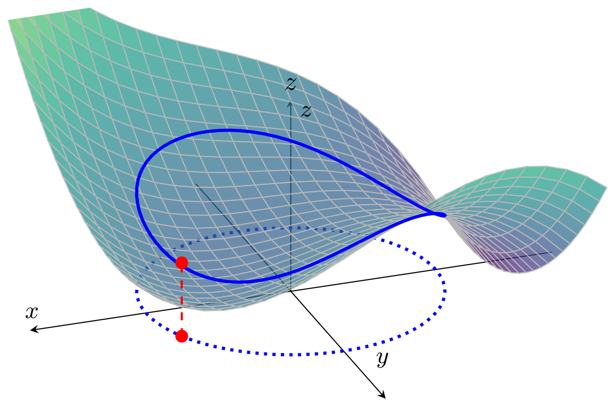


Figure 11: An example of a function $f : \mathbb{R}^2 \longrightarrow \mathbb{R}$ constrained to the unit circle, where $g(x, y) = x^2 + y^2 - 1 = 0$. The blue curve in the xy-plane is the constraint set, and the upper blue curve is its image on the graph of f.

To solve this constraint problem, we use the method of Lagrange multipliers. The basic idea is to convert a constrained problem into a form such that the derivative test of an unconstrained problem can still be applied.

Theorem 1.18 (Lagrange Multipliers Theorem)

Let $f : \mathbb{R}^n \to \mathbb{R}$ and $g : \mathbb{R}^n \to \mathbb{R}^c$ be C^1 functions, and define $M = \{x \in \mathbb{R}^n : g(x) = 0\}$. Suppose that

1. $a \in M$ is a local maximum of f restricted to M,
2. $\text{rank}(Dg_a) = c$.

Then there exists a unique row vector $\lambda \in \mathbb{R}^{1 \times c}$, called the **Lagrange multiplier vector**, such that

$$Df_a = \lambda Dg_a. \tag{294}$$

Proof. Since $\text{rank}(Dg_a) = c$, the $c \times n$ matrix Dg_a contains an invertible $c \times c$ submatrix. After reordering the coordinates of $\mathbb{R}^n$ if necessary, write

$$x = (u, v) \in \mathbb{R}^{n-c} \oplus \mathbb{R}^c, \tag{295}$$

Then, we can write $a = (\alpha, \beta)$, so that $D_v g_{(\alpha,\beta)}$ is invertible. Since $a \in M$, we have $g(\alpha, \beta) = g(a) = 0$. Therefore, by the implicit function theorem, there exist open neighborhoods $U \ni \alpha$ and $V \ni \beta$, together with a unique C^1 function

$$h : U \to V, \tag{296}$$

such that $h(\alpha) = \beta$ and, for every $(u, v) \in U \times V$, $g(u, v) = 0 \iff v = h(u)$. Moreover, for all $u \in U$,

$$Dh_u = -\big(D_v g_{(u,h(u))}\big)^{-1} D_u g_{(u,h(u))}. \tag{297}$$

[1] In physics, these types of "well-behaved" constraints are known as *holonomic constraints*.

and in particular, $Dh_\alpha = -\big(D_v g_a\big)^{-1} D_u g_a$. Define the restriction of f to this local parameterization of M by

$$\varphi : U \to \mathbb{R}, \qquad \varphi(u) = f(u, h(u)). \tag{298}$$

Since $g(u, h(u)) = 0$, every point $(u, h(u))$ lies in M. Furthermore,

$$(\alpha, h(\alpha)) = (\alpha, \beta) = a. \tag{299}$$

Because a is a local maximum of f restricted to M, after shrinking U if necessary, α is a local maximum of φ. Therefore,

$$D\varphi_\alpha = 0. \tag{300}$$

By the chain rule, we have $D\varphi_\alpha = D_u f_a + D_v f_a Dh_\alpha$, and substituting the formula for Dh_α gives

$$0 = D_u f_a - D_v f_a \big(D_v g_a\big)^{-1} D_u g_a \implies D_u f_a = D_v f_a \big(D_v g_a\big)^{-1} D_u g_a. \tag{301}$$

If we define $\lambda := D_v f_a \big(D_v g_a\big)^{-1} \in \mathbb{R}^{1 \times c}$, then we can see that

1. From above, we have $D_u f_a = \lambda D_u g_a$.
2. By the definition of λ, $\lambda D_v g_a = D_v f_a \big(D_v g_a\big)^{-1} D_v g_a = D_v f_a$.

Combining these two into matrix form, we have

$$Df_a = \begin{pmatrix} D_u f_a & D_v f_a \end{pmatrix} = \begin{pmatrix} \lambda D_u g_a & \lambda D_v g_a \end{pmatrix} = \lambda \begin{pmatrix} D_u g_a & D_v g_a \end{pmatrix} = \lambda D g_a. \tag{302}$$

It remains to prove uniqueness. Suppose that $\lambda, \widetilde{\lambda} \in \mathbb{R}^{1 \times c}$ both satisfy

$$Df_a = \lambda D g_a = \widetilde{\lambda} D g_a \implies (\lambda - \widetilde{\lambda}) D g_a = 0. \tag{303}$$

Restricting this equality to the v-columns gives $(\lambda - \widetilde{\lambda}) D_v g_a = 0$. Since $D_v g_a$ is invertible, it follows that $\lambda - \widetilde{\lambda} = 0$, and we are done.

By transposing Equation 294, we can get a matrix realization of

$$\begin{pmatrix} \frac{\partial f}{\partial x_1}(a) \\ \vdots \\ \frac{\partial f}{\partial x_n}(a) \end{pmatrix} = \begin{pmatrix} \frac{\partial g_1}{\partial x_1}(a) & \dots & \frac{\partial g_c}{\partial x_1}(a) \\ \vdots & \ddots & \vdots \\ \frac{\partial g_1}{\partial x_n}(a) & \dots & \frac{\partial g_c}{\partial x_n}(a) \end{pmatrix} \begin{pmatrix} \lambda_1 \\ \vdots \\ \lambda_c \end{pmatrix} \tag{304}$$

$$= \lambda_1 \begin{pmatrix} \frac{\partial g_1}{\partial x_1}(a) \\ \vdots \\ \frac{\partial g_1}{\partial x_n}(a) \end{pmatrix} + \lambda_2 \begin{pmatrix} \frac{\partial g_2}{\partial x_1}(a) \\ \vdots \\ \frac{\partial g_2}{\partial x_n}(a) \end{pmatrix} + \dots + \lambda_c \begin{pmatrix} \frac{\partial g_c}{\partial x_1}(a) \\ \vdots \\ \frac{\partial g_c}{\partial x_n}(a) \end{pmatrix} \tag{305}$$

This equation tells us that at any critical points a of f evaluated under the equality constraints, the total derivative of f at a can be expressed as a unique linear combination of the total derivatives of the constraints $D(g_i)_a$, with the Lagrange multipliers acting as coefficients. Therefore, finding the critical points a of f constrained with g is equivalent to solving the system of $c + n$ equations for the n unknowns in x and c unknowns in λ.

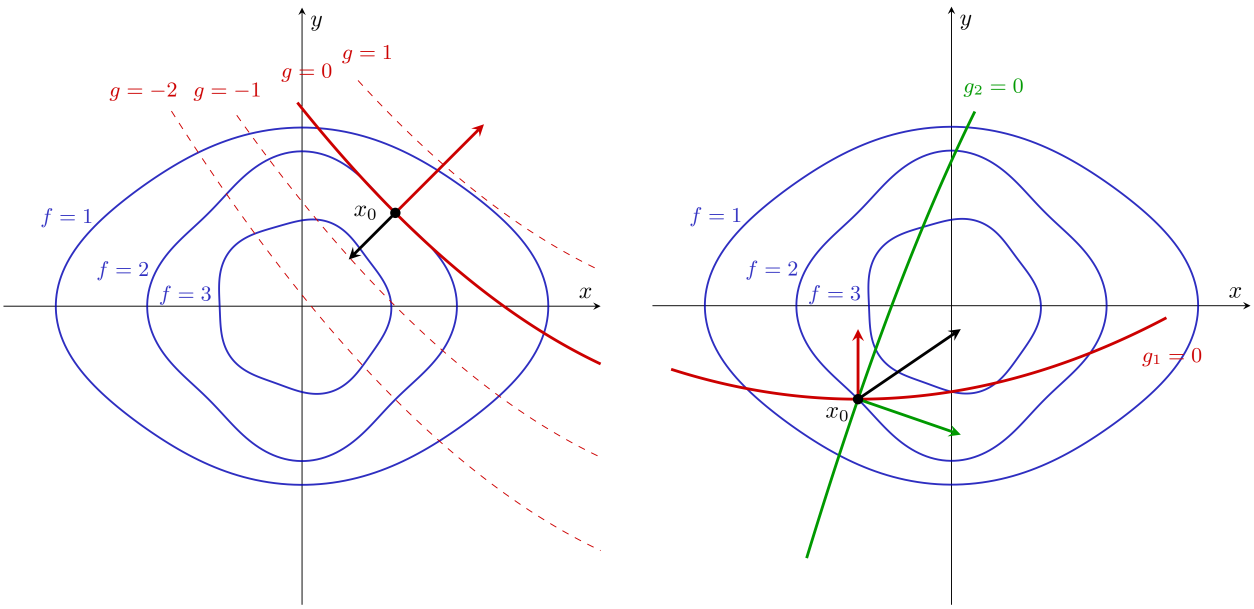
$$g(x) = 0 \iff \begin{cases} g_1(x) & = 0 \\ \dots & = 0 \\ g_c(x) & = 0 \end{cases} \tag{306}$$

$$Df_a = \lambda D g_a \iff \begin{cases} \frac{\partial f}{\partial x_1}(a) & = \lambda_1^* \frac{\partial g_1}{\partial x_1}(a) + \lambda_2^* \frac{\partial g_2}{\partial x_1}(a) + \dots + \lambda_c^* \frac{\partial g_c}{\partial x_1}(a) \\ \dots & = \dots \\ \frac{\partial f}{\partial x_n}(a) & = \lambda_1^* \frac{\partial g_1}{\partial x_n}(a) + \lambda_2^* \frac{\partial g_2}{\partial x_n}(a) + \dots + \lambda_c^* \frac{\partial g_c}{\partial x_n}(a) \end{cases} \tag{307}$$

This relationship between the gradient of the function and gradients of the constraints rather naturally leads to a reformulation of the original problem, known as the *Lagrangian function.* That is, in order to find the maximum/minimum of f subjected to the equality constraint $g(x) = 0$, we form the Lagrangian function

$$\mathcal{L}(x, \lambda) = f(x) - \lambda^T g(x) \tag{308}$$

and find the stationary points of $\mathcal{L}$ considered as a function of $x \in \mathbb{R}^n$ and the Lagrange multiplier $\lambda \in \mathbb{R}^c$. The main advantage to this method is that it allows the optimization to be solved without explicit parameterization in terms of the constraints.



(a) When there is a single constraint $g : \mathbb{R}^n \longrightarrow \mathbb{R}$, $\nabla f(x_0)$ can be expressed as a linear combination of $\nabla g(x_0)$.

(b) When there are multiple constraints g_1, g_2, $\nabla f(x_0)$ can be expressed as a linear combination of $\nabla g_1(x_0)$ and $\nabla g_2(x_0)$.

Figure 12: From the properties of the gradient, $\nabla f(x_0)$ is orthogonal to the level set of points satisfying $f(x) = f(x_0)$ at point x_0. We can see that the point where the contour line of $g(x) = 0$ tangentially touches the contours of f is the maximum. Since it intersects it tangentially, the gradient vector at that point $\nabla g(x_0)$ is parallel to $\nabla f(x_0)$.

From the properties of the gradient introduced before, $\nabla f(x_0)$ is orthogonal to the level set of points satisfying $f(x) = c$ at the point x_0. But this level set $f(x) = c$ actually intersects the level set determined by $g(x) = c$ at the point x_0 and is indistinguishable from each other at x_0. This means that $\nabla g(x_0)$ is normal the level set of $g(x) = c$ at x_0 $\iff$ it is normal to the level set of $f(x) = c$ at x_0. But $\nabla f(x_0)$ is also normal at that point, so $\nabla f(x_0)$ must be parallel to $\nabla g(x_0)$.

Example 1.23 (1 Constraint in $\mathbb{R}^2$)

Find the extrema of f on the unit circle, where

$$f(x, y) = xy, \qquad g(x, y) = x^2 + y^2 - 1. \tag{309}$$

Since $Dg_{(x,y)}$ is nonzero on the constraint set, the Lagrange multiplier equation is

$$\begin{aligned} Df_{(x,y)} &= \lambda Dg_{(x,y)}, \\ \begin{pmatrix} y & x \end{pmatrix} &= \lambda \begin{pmatrix} 2x & 2y \end{pmatrix}, \qquad x^2 + y^2 = 1. \end{aligned} \tag{310}$$

The resulting candidates and their values are

$$\begin{array}{c|c} (x,y) & f(x,y) \\ \hline \left(\pm\frac{1}{\sqrt{2}}, \pm\frac{1}{\sqrt{2}}\right) & \frac{1}{2} \\ \left(\pm\frac{1}{\sqrt{2}}, \mp\frac{1}{\sqrt{2}}\right) & -\frac{1}{2}. \end{array} \tag{311}$$

Thus the maximum is $1/2$ and the minimum is $-1/2$.

Example 1.24 (1 Constraint in $\mathbb{R}^3$)

Find the extrema of f on the unit sphere, where

$$f(x,y,z) = x + y + z, \qquad g(x,y,z) = x^2 + y^2 + z^2 - 1. \tag{312}$$

Since $Dg_{(x,y,z)}$ is nonzero on the constraint set, the equation $Df_a = \lambda Dg_a$ gives

$$\begin{pmatrix} 1 & 1 & 1 \end{pmatrix} = \lambda \begin{pmatrix} 2x & 2y & 2z \end{pmatrix}, \qquad x^2 + y^2 + z^2 = 1. \tag{313}$$

Hence $x = y = z$, so the only candidates are

$$\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right) \qquad \text{and} \qquad \left(-\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}\right). \tag{314}$$

Their respective values are $\sqrt{3}$ and $-\sqrt{3}$, giving the maximum and minimum.

Example 1.25 (2 Constraints in $\mathbb{R}^3$)

Find the extrema of f on the intersection of the unit sphere and the plane $x + y + z = 0$, where

$$f(x,y,z) = x + 2y + 3z, \qquad g(x,y,z) = \begin{pmatrix} x^2 + y^2 + z^2 - 1 \\ x + y + z \end{pmatrix}. \tag{315}$$

The rows of $Dg_{(x,y,z)}$ are independent on the constraint set. Writing $\lambda = \begin{pmatrix} \lambda_1 & \lambda_2 \end{pmatrix}$, the Lagrange multiplier equation becomes

$$\begin{gathered} \begin{pmatrix} 1 & 2 & 3 \end{pmatrix} = \begin{pmatrix} \lambda_1 & \lambda_2 \end{pmatrix} \begin{pmatrix} 2x & 2y & 2z \\ 1 & 1 & 1 \end{pmatrix}, \\ x^2 + y^2 + z^2 = 1, \qquad x + y + z = 0. \end{gathered} \tag{316}$$

Adding the three components of the first equation gives $\lambda_2 = 2$, and therefore

$$\begin{pmatrix} -1 & 0 & 1 \end{pmatrix} = 2\lambda_1 \begin{pmatrix} x & y & z \end{pmatrix}. \tag{317}$$

Thus the candidates are

$$\left(-\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right) \qquad \text{and} \qquad \left(\frac{1}{\sqrt{2}}, 0, -\frac{1}{\sqrt{2}}\right), \tag{318}$$

with respective values $\sqrt{2}$ and $-\sqrt{2}$. Therefore these are the maximum and minimum, respectively.

1.9 Exercises

Exercise 1.1 (Rudin 9.6)

If $f(0,0) = 0$ and
$$f(x,y) = \frac{xy}{x^2+y^2} \qquad \text{if } (x,y) \neq (0,0), \tag{319}$$
prove that $(D_1 f)(x,y)$ and $(D_2 f)(x,y)$ exist at every point of $\mathbb{R}^2$, although f is not continuous at $(0,0)$.

Solution. Easy.

Exercise 1.2 (Rudin 9.7)

Suppose that f is a real-valued function defined in an open set $E \subset \mathbb{R}^n$, and that the partial derivatives $D_1 f, \ldots, D_n f$ are bounded in E. Prove that f is continuous in E.

Solution. Proved in this theorem. If we had stronger assumptions like the function was differentiable over convex set, we can say that since the partials are bounded, the total derivative Df is bounded—by, say M—over E. If E is convex, then by MVT, we have $f(y) - f(x) \leq M\|x-y\|$, and so f is uniformly continuous.

Exercise 1.3 (Rudin 9.8)

Suppose that f is a differentiable real function in an open set $E \subset \mathbb{R}^n$, and that f has a local maximum at a point $x \in E$. Prove that
$$f'(x) = 0. \tag{320}$$

Solution. Proved in first derivative test.

Exercise 1.4 (Rudin 9.9)

If f is a differentiable mapping of a connected open set $E \subset \mathbb{R}^n$ into $\mathbb{R}^m$, and if $f'(x) = 0$ for every $x \in E$, prove that f is constant in E.

Solution. It would be nice if we can upgrade connectedness to path connectedness since then we can just work with path functions. Luckily, open sets in $\mathbb{R}^n$ are locally path connected, so they are also path connected. Therefore, for any $x, y \in E$, find a path $\gamma : [0,1] \to E$ s.t. $\gamma(0) = x, \gamma(1) = y$. Then, $(f \circ \gamma)'(t) = Df_{\gamma(t)} \gamma'(t) = 0\gamma'(t) = 0$.[a] So, by the univariate MVT applied to each component, $(f \circ \gamma)(t) = c$ for some constant c. γ was chosen with arbitrary $x, y \in E$, so this implies that $f(x) = c$.

[a] Technically, γ must be differentiable, but we can just say that γ is piecewise linear, aka polygonally connected.

Exercise 1.5 (Rudin 9.10)

If f is a real function defined in a convex open set $E \subset \mathbb{R}^n$, such that $(D_1 f)(x) = 0$ for every $x \in E$, prove that $f(x)$ depends only on $x_2, \ldots, x_n$.
Show that the convexity of E can be replaced by a weaker condition, but that some condition is required. For example, if $n = 2$ and E is shaped like a horseshoe, the statement may be false.

Exercise 1.6 (Rudin 9.11)

If f and g are differentiable real functions in $\mathbb{R}^n$, prove that

$$\nabla(fg) = f\nabla g + g\nabla f, \tag{321}$$

and that

$$\nabla\left(\frac{1}{f}\right) = -f^{-2}\nabla f \tag{322}$$

wherever $f \neq 0$.

Solution. In the book, Rudin defines for $f : \mathbb{R}^n \to \mathbb{R}$,

$$\nabla f(x) = \sum_{i=1}^n \frac{\partial f}{\partial x_i}(x) \cdot e_i \tag{323}$$

Since a partial derivative is just a univariate derivative, we can invoke the same product/quotient rules of differentiation.

$$\begin{aligned}
\big(\nabla(fg)\big)(x) &= \sum_{i=1}^n \left(\frac{\partial(fg)}{\partial x_i}\right)(x) \cdot e_i && (324)\\
&= \sum_{i=1}^n \left(\Big(\frac{\partial f}{\partial x_i}\Big) g + f\Big(\frac{\partial g}{\partial x_i}\Big)\right)(x) \cdot e_i && (325)\\
&= \sum_{i=1}^n \left(\Big(\frac{\partial f}{\partial x_i}\Big)(x)\, g(x) + f(x)\Big(\frac{\partial g}{\partial x_i}\Big)(x)\right) \cdot e_i && (326)\\
&= \nabla f(x) g(x) + f(x) \nabla g(x) && (327)
\end{aligned}$$

The same logic for inverse:

$$\nabla(f^{-1})(x) = \sum_{i=1}^n \frac{\partial f^{-1}}{\partial x_i}(x) \cdot e_i = \sum_{i=1}^n \frac{-(\partial f/\partial x_i)(x)}{(f(x))^2} \cdot e_i = \frac{-1}{(f(x))^2} \nabla f(x) \tag{328}$$

Exercise 1.7 (Rudin 9.12)

Fix two real numbers a and b, $0 < a < b$. Define a mapping $f = (f_1, f_2, f_3)$ of $\mathbb{R}^2$ into $\mathbb{R}^3$ by

$$\begin{aligned}
f_1(s,t) &= (b + a\cos s)\cos t, && (329)\\
f_2(s,t) &= (b + a\cos s)\sin t, && (330)\\
f_3(s,t) &= a \sin s. && (331)
\end{aligned}$$

Describe the range K of f. (It is a certain compact subset of $\mathbb{R}^3$.)

(a) Show that there are exactly 4 points $p \in K$ such that

$$(\nabla f_1)(f^{-1}(p)) = 0. \tag{332}$$

Find these points.

(b) Determine the set of all $q \in K$ such that

$$(\nabla f_3)(f^{-1}(q)) = 0. \tag{333}$$

(c) Show that one of the points p found in part (a) corresponds to a local maximum of f_1, one corresponds to a local minimum, and that the other two are neither (they are so-called "saddle points").

Which of the points q found in part (b) correspond to maxima or minima?

(d) Let λ be an irrational real number, and define $g(t) = f(t, \lambda t)$. Prove that g is a one-to-one mapping of $\mathbb{R}^1$ onto a dense subset of K. Prove that

$$|g'(t)|^2 = a^2 + \lambda^2(b + a\cos t)^2. \tag{334}$$

Solution. The range is a torus, and you can see it by decomposing the parameterizations[a] as

$$f(s,t) = \begin{pmatrix} b\cos(t) + a\cos(s)\cos(t) \\ b\cos(t) + a\cos(s)\sin(t) \\ a\sin(s) \end{pmatrix} \tag{335}$$

$$= \begin{pmatrix} b\cos(t) \\ b\sin(t) \\ 0 \end{pmatrix} + a\left\{ \cos(s)\begin{pmatrix} \cos(t) \\ \sin(t) \\ 0 \end{pmatrix} + \sin(s)\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\} \tag{336}$$

1. We can see that

$$\nabla f_1(s,t) = \begin{pmatrix} -a\sin(s)\cos(t) \\ -(b + a\cos(s))\sin(t) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \tag{337}$$

Since $a < b \implies b + a\cos(s) \neq 0$ for all s, this implies that $\sin(t) = 0$ and $\sin(s) = 0$. Therefore, the points $(s,t) = \{(0,0), (0,\pi), (\pi,0), (\pi,\pi)\}$ satisfy the equation, and their images in K are

$$f(0,0) = \begin{pmatrix} b+a \\ 0 \\ 0 \end{pmatrix}, f(0,\pi) = \begin{pmatrix} -b-a \\ 0 \\ 0 \end{pmatrix}, f(\pi,0) = \begin{pmatrix} b-a \\ 0 \\ 0 \end{pmatrix}, f(\pi,\pi) = \begin{pmatrix} -b+a \\ 0 \\ 0 \end{pmatrix} \tag{338}$$

2. We see that

$$\nabla f_3(s,t) = \begin{pmatrix} a\cos(s) \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \tag{339}$$

which implies that $\cos(s) = 0$, which implies that $s = \frac{\pi}{2}, \frac{3\pi}{2}$. Therefore, the zeros are $\{(s,t) \in \mathbb{R}^2 \colon s = \frac{\pi}{2}, \frac{3\pi}{2}\}$ and its image is the set of all points of either of the following forms

$$\begin{pmatrix} b\cos(t) \\ b\sin(t) \\ a \end{pmatrix}, \qquad \begin{pmatrix} b\cos(t) \\ b\sin(t) \\ -a \end{pmatrix} \tag{340}$$

3. Since $-1 \leq \sin(x), \cos(x) \leq 1$, it is clear that $-a - b \leq f_1(s,t) \leq a + b$. $(0,0)$ attains this max, $(0,\pi)$ attains the minimum, and so the final two must be saddle points. Indeed, for $(s,t) = (\pi, 0)$, if we perturb the t, $f_1(s,t)$ decreases, and if we perturb the s, $f_1(s,t)$ increases. For $(s,t) = (0,\pi)$ if we perturb the t, $f_1(s,t)$ increases, and if we perturb the t, $f_1(s,t)$ decreases.

4. It suffices to show that the map to the quotient space $t \mapsto (t, \lambda t) \in \frac{[0,2\pi)^2}{\sim}$ is injective, where $(a,b) \sim (c,d)$ if $a = c + 2\pi k_1$ and $b = d + 2\pi k_2$ for some $k_1, k_2 \in \mathbb{Z}$. Now let $t_1, t_2 \in \mathbb{R}$ with $g(t_1) = g(t_2)$, i.e. $(t_1, \lambda t_1) \sim (t_2, \lambda t_2)$. By looking at the first coordiante, this means that $t_1 = t_2 + 2\pi k_1$ for some $k_1 \in \mathbb{Z}$. By multiplying by λ, we have $\lambda t_1 = \lambda t_2 + 2\pi k_1 \lambda$, but from the second coordinate, this means that $k_1\lambda \in \mathbb{Z}$. If $k_1 \neq 0$, then $\lambda \in \mathbb{Q}$, a contradiction. So $k_1 = 0$, which implies that $t_1 = t_2$.

   Now we show that $t \to (t, \lambda t)$ has a dense image. Let $(a,b) \in \frac{[0,2\pi]^2}{\sim}$ and fix $\epsilon > 0$. Then, let

$$t \in \bigcup_{k\in\mathbb{Z}} (2\pi k + a - \epsilon, 2\pi k + a + \epsilon) \tag{341}$$

   Then, we wish to show that

$$\lambda t \in \bigcup_{k\in\mathbb{Z}} (2\pi k\lambda + \lambda a - \lambda\epsilon, 2\pi k\lambda + \lambda a + \lambda\epsilon) \tag{342}$$

which is true since the image of $k \to \{2\pi\lambda k\}$ is dense in $[0, 2\pi]$.
To verify the norm, I just did it manually by finding

$$g'(t) = \begin{pmatrix} -\lambda b \sin(\lambda t) - a\lambda \cos(t)\sin(\lambda t) - a\sin(t)\cos(\lambda t) \\ \lambda b\cos(\lambda t) + a\lambda\cos(t)\cos(\lambda t) - a\sin(t)\sin(\lambda t) \\ a\cos(t) \end{pmatrix} \tag{343}$$

After some tedious calculations, I got the identity.

[a]Very good video here: https://www.youtube.com/watch?v=heFhbV5J_bY.

Exercise 1.8 (Rudin 9.13)

Suppose that f is a differentiable mapping of $\mathbb{R}^1$ into $\mathbb{R}^3$ such that $|f(t)| = 1$ for every t. Prove that

$$f'(t) \cdot f(t) = 0. \tag{344}$$

Interpret this result geometrically.

Solution. We must have $\|f(t)\|^2 = 1$. By differentiating both sides, we get $2f(t) \cdot f(t) = 0$, which is the desired result. This means that if f lives in the unit sphere, then the direction of its change must always be tangent to the sphere.

Exercise 1.9 (Rudin 9.14)

Define $f(0,0) = 0$ and

$$f(x,y) = \frac{x^3}{x^2+y^2} \qquad \text{if } (x,y) \neq (0,0). \tag{345}$$

(a) Prove that $D_1 f$ and $D_2 f$ are bounded functions in $\mathbb{R}^2$. (Hence f is continuous.)
(b) Let u be any unit vector in $\mathbb{R}^2$. Show that the directional derivative $(D_u f)(0,0)$ exists, and that its absolute value is at most 1.
(c) Let γ be a differentiable mapping of $\mathbb{R}^1$ into $\mathbb{R}^2$ (in other words, γ is a differentiable curve in $\mathbb{R}^2$), with $\gamma(0) = (0,0)$ and $|\gamma'(0)| > 0$. Put $g(t) = f(\gamma(t))$ and prove that g is differentiable for every $t \in \mathbb{R}^1$.
If $\gamma \in \mathscr{C}'$, prove that $g \in \mathscr{C}'$.
(d) In spite of this, prove that f is not differentiable at $(0,0)$.
Hint: Formula (40) fails.

Solution. Listed.

1. When $(x,y) \neq (0,0)$,

$$\frac{\partial}{\partial x}\frac{x^3}{x^2+y^2} = \frac{3x^2(x^2+y^2) - 2x^4}{x^2+y^2} = \frac{x^4+3x^2y^2}{x^4+2x^2y^2+y^4} < 1 + \frac{3}{2} \tag{346}$$

$$\frac{\partial}{\partial y}\frac{x^3}{x^2+y^2} = -\frac{2x^3y}{(x^2+y^2)^2} \tag{347}$$

We can see that if $|x| > |y|$, then $|x^3y| < x^4 \leq (x^2+y^2)^2$, and if $|x| \leq |y|$, then $|x^3y| \leq y^4 \leq$

$(x^2 + y^2)^2$. So both partials are bounded. Hence, it is continuous. At $(0,0)$, the partials are

$$\frac{\partial f}{\partial x}(0,0) = \lim_{h \to 0} \frac{f(h,0) - f(0,0)}{h} = \frac{h^3}{h \cdot (h^2 + 0^2)} = 1 \tag{348}$$

$$\frac{\partial f}{\partial y}(0,0) = \lim_{h \to 0} \frac{f(0,h) - f(0,0)}{h} = 0 \tag{349}$$

2. Let $u = (u_1, u_2)$ be any unit vector. Then,

$$D_u f(0,0) = \lim_{h \to 0} \frac{f(hu_1, hu_2) - f(0,0)}{h} \tag{350}$$

$$= \lim_{h \to 0} \frac{1}{h} \frac{h^3 u_1^3}{h^2 u_1^2 + h^2 u_2^2} \tag{351}$$

$$= \frac{u_1^3}{u_1^2 + u_2^2} = u_1^3 \tag{352}$$

and since $|u_1| \leq 1$, we have $|D_u f(0,0)| = |u_1^3| \leq 1$.

3. When $\gamma(t) \neq 0$, f is differentiable at $\gamma(t)$, so we can simply use chain rule. Let $t_0 \in \mathbb{R}$ with $\gamma(t_0) = (0,0)$. Since γ is differentiable everywhere, we can take the first order approximation

$$\gamma(t_0 + h) = \underbrace{\gamma(t_0)}_{=(0,0)} + \gamma'(t_0)h + o(h) \tag{353}$$

Noticing the homogeneity property $f(\lambda x, \lambda y) = \lambda f(x,y)$ is very important, since we can do

$$\frac{f(\gamma(t_0 + h) - f(\gamma(t_0))}{h} = \frac{1}{h} f(\gamma(t_0 + h)) - \frac{1}{h} f(\gamma(t_0)) \tag{354}$$

$$= f\bigg(\frac{\gamma(t_0 + h)}{h}\bigg) \tag{355}$$

$$= f\bigg(\frac{\gamma'(t_0)h + o(h)}{h}\bigg) \tag{356}$$

$$= f\big(\gamma'(t_0) + \frac{o(h)}{h}\big) \tag{357}$$

So, by taking the limit and from continuity of f, we have

$$\lim_{h \to 0} \frac{f(\gamma(t_0 + h) - f(\gamma(t_0)))}{h} = \lim_{h \to 0} f\big(\gamma'(t_0) + \frac{o(h)}{h}\big) = f\big(\gamma'(t_0) + \underbrace{\lim_{h \to 0} \frac{o(h)}{h}}_{=0}\big) = f\big(\gamma'(t_0)\big) \tag{358}$$

To prove that it is C^1, we know that f is C^1 outside the origin, so the composition $g = f \circ \gamma$ is C^1 for $t \in \mathbb{R}$ where $\gamma(t) \neq 0$. The hard part is determining C^1 for $t_0 \in \mathbb{R}$ where $\gamma(t_0) = 0$. We know that

$$g'(t) = \begin{cases} Df(\gamma(t))\gamma'(t) & \text{if } \gamma(t) \neq 0 \\ f(\gamma'(t)) & \text{if } \gamma(t) = 0 \end{cases} \tag{359}$$

After some unsuccessful attempts,[a] GPT recommended me this solution. Let $t_0 \in \mathbb{R}$ such that $\gamma(t_0) = 0$. First, note that

(a) the partials are bounded, so $|Df(\gamma(t))\gamma'(t)| \leq M|\gamma'(t)|$ for some M.

(b) $f(x,y)$ as a function is bounded by $|f(x,y)| \leq \frac{x^3}{x^2+y^2} \leq |x| \leq |(x,y)|$, so $|f(\gamma'(t))| \leq |\gamma'(t)|$

Therefore, there exists a constant $C = \max\{M, 1\}$ where $|g'(t)| \leq C|\gamma'(t)|$. We can consider two cases.

(a) $\gamma'(t_0) = 0$. This is easy, since by C^1 and our assumption, we have

$$0 \leq \lim_{t \to t_0} |g'(t)| \leq \lim_{t \to t_0} C|\gamma'(t)| = \gamma'(t_0) = 0 \implies \lim_{t \to t_0} g'(t) = 0 = f(0) = f(\gamma'(t_0)) = g'(t_0) \tag{360}$$

(b) $\gamma'(t_0) \neq 0$. $\gamma'(t_0)$. We want to do something like compute

$$\lim_{t \to t_0} g'(t) = \lim_{t \to t_0} Df_{\gamma(t)} \gamma'(t) \tag{361}$$

However, the fact that the derivative $Df_{\gamma(t)}$ may not be continuous at $\gamma(t_0) = 0$ prevents us from going further. We want to compute this in a roundabout way. First, let's define the quotient

$$v(t) = \frac{\gamma(t) - \gamma(t_0)}{t - t_0} = \frac{\gamma(t)}{t - t_0} \tag{362}$$

Since γ is differentiable at t_0, we have $\lim_{t \to t_0} v(t) = \gamma'(t_0)$. Because $\gamma'(t_0) \neq 0$, we have $v(t) \neq 0$ for some neighborhood of t_0. Therefore,

$$\gamma(t) = (t - t_0) v(t) \neq 0 \tag{363}$$

for all such t. It is easy to verify that since $f(\lambda x) = \lambda f(x)$, then $Df_{\lambda x} = Df_x$. Therefore, for all t in the mentioned neighborhood of t_0, we have

$$Df_{\gamma(t)} = Df_{(t - t_0) v(t)} = Df_{v(t)} \tag{364}$$

Now, we can rewrite the chain rule g' as

$$g'(t) = Df_{v(t)}(\gamma'(t)) \tag{365}$$

We like this form much better since now, $x \to Df_x$ is continuous away from the origin, so *now* we can use product rule to split the limits

$$\begin{aligned} \lim_{t \to t_0} g'(t) &= \lim_{t \to t_0} Df_{v(t)} \gamma'(t) && (366) \\ &= \big(\lim_{t \to t_0} Df_{v(t)} \big) \big(\lim_{t \to t_0} \gamma'(t) \big) && (367) \\ &= Df_{\gamma'(t_0)} \gamma'(t_0) && (368) \end{aligned}$$

Now this is where the homogeneity property is essential again. Consider $g(\lambda) = \lambda f(x) = f(\lambda x)$. Differentiating both sides at $\lambda = 1$ gives

$$g'(1) = f(x) = Df_{\lambda x} x = Df_x x \tag{369}$$

where the final step was shown earlier. Therefore,

$$\lim_{t \to t_0} g'(t) = Df_{\gamma'(t_0)} \gamma'(t_0) = f(\gamma'(t_0)) = g'(t_0) \tag{370}$$

and therefore g is C^1 at t_0.

4. To prove that it is not differentiable, let's take the directional derivative in direction vector $u = (\frac{1}{2}, \frac{\sqrt{3}}{2})$. According to the formula in (2), we have

$$D_u f(0,0) = u_1^3 = \frac{1}{8} \tag{371}$$

But using a linear combination of the partials, we have

$$\begin{pmatrix} \frac{\partial f}{\partial x}(0,0) & \frac{\partial f}{\partial y}(0,0) \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} \frac{1}{2} \\ \frac{\sqrt{3}}{2} \end{pmatrix} = \frac{1}{2} \tag{372}$$

which is not equal. Therefore, f is not differentiable at $(0,0)$.

[a] I first tried to split the limits as $\lim_{t \to t_0} g'(t) = \lim_{t \to t_0} Df(\gamma(t)) \gamma'(t) = \big(\lim_{t \to t_0} Df(\gamma(t)) \big) \big(\lim_{t \to t_0} \gamma'(t) \big) = \big(\lim_{t \to t_0} Df(\gamma(t)) \big) \gamma'(t_0)$, but I can't do this since Df may not be continuous at $\gamma(t_0)$, so we have no guarantee that $\lim_{t \to t_0} Df(\gamma(t))$ exists.

Exercise 1.10 (Rudin 9.15)

Define $f(0,0) = 0$, and put

$$f(x,y) = x^2 + y^2 - 2x^2y - \frac{4x^6y^2}{(x^4+y^2)^2} \tag{373}$$

if $(x,y) \neq (0,0)$.

(a) Prove, for all $(x,y) \in \mathbb{R}^2$, that

$$4x^4y^2 \leq (x^4+y^2)^2. \tag{374}$$

Conclude that f is continuous.

(b) For $0 \leq \theta \leq 2\pi$, $-\infty < t < \infty$, define

$$g_\theta(t) = f(t\cos\theta, t\sin\theta). \tag{375}$$

Show that

$$g_\theta(0) = 0, \qquad g_\theta'(0) = 0, \qquad g_\theta''(0) = 2. \tag{376}$$

Each g_θ has therefore a strict local minimum at $t = 0$.
In other words, the restriction of f to each line through $(0,0)$ has a strict local minimum at $(0,0)$.

(c) Show that $(0,0)$ is nevertheless not a local minimum for f, since

$$f(x,x^2) = -x^4. \tag{377}$$

Solution. Listed.

1. We know that $0 \leq (x^4-y^2)^2 = x^8 - 2x^4y^2 + y^4$. Therefore, we can see that

$$|f(x,y)| = \left| x^2 + y^2 - 2x^2y - x^2\frac{4x^4y^2}{(x^4+y^2)^2} \right| \tag{378}$$

$$\leq |x^2+y^2-2x^2y| + x^2 \underbrace{\left|\frac{4x^4y^2}{(x^4+y^2)^2}\right|}_{<1} \tag{379}$$

$$\leq |x^2+y^2-2x^2y| + x^2 \tag{380}$$

$$\leq 2x^2 + y^2 + 2x^2|y| \tag{381}$$

So let's bound the supremum norm by $\delta = \min\{\sqrt{\frac{\epsilon}{5}}, \sqrt[3]{\frac{\epsilon}{5}}\}$. Then,

$$|x|,|y| < \delta \implies |f(x,y)| \leq 2x^2 + y^2 + 2x^2|y| \leq 2\sqrt{\frac{\epsilon}{5}}^2 + \sqrt{\frac{\epsilon}{5}}^2 + 2\sqrt[3]{\frac{\epsilon}{5}}^3 = \epsilon \tag{382}$$

2. Let's substitute $a = \cos\theta, b = \sin\theta$, and then write

$$g_\theta(t) = f(at,bt) = t^2 - 2a^2bt^3 - \frac{4a^6b^2t^4}{(a^4t^2+b^2)^2} \tag{383}$$

Also, after some tedious calculations,

$$g_\theta'(t) = 2t - 6a^2bt^2 - \frac{16a^6b^4t^3}{(a^4t^2+b^2)^3} \tag{384}$$

Again, we differentiate to get

$$g_\theta''(t) = 2 - 12a^2bt + \frac{48a^6b^4t^2(a^4t^2-b)}{(a^4t^2+b^2)^4} \tag{385}$$

From this, we can see that $g_\theta(0) = 0, g_\theta'(0) = 0, g_\theta''(0) = 2$. Therefore, by the second derivative test, the restriction of g to each line passing through $(0,0)$ has a strict local minimum at $t = 0$.

3. It is easy to check
$$f(x, x^2) = x^2 + x^4 - 2x^4 - \frac{4x^10}{4x^8} = -x^4 \tag{386}$$
Therefore, for any open neighborhood U of $(0,0)$, we can find an $\epsilon > 0$ so small such that $(\epsilon, \epsilon^2) \in U$ and $f(\epsilon, \epsilon^2) = -\epsilon^4 < 0 = f(0,0)$.

Exercise 1.11 (Rudin 9.16)

Show that the continuity of f' at the point a is needed in the inverse function theorem, when the case $n = 1$: If
$$f(t) = t + 2t^2 \sin\left(\frac{1}{t}\right) \tag{387}$$
for $t \neq 0$, and $f(0) = 0$, then $f'(0) = 1$, f' is bounded in $(-1,1)$, but f is not one-to-one in any neighborhood of 0.

Solution. For $t \neq 0$, we have $f'(t) = 1 + 4t\sin\left(\frac{1}{t}\right) - 2\cos\left(\frac{1}{t}\right)$. When $t = 0$, we use the limit definition
$$f'(0) = \lim_{h\to 0} \frac{f(0+h) - f(0)}{h} = \lim_{h \to 0} \frac{f(h)}{h} \tag{388}$$
$$= \lim_{h\to 0} \frac{1}{h}\big(h + 2h^2 \sin(1/h)\big) \tag{389}$$
$$= \lim_{h \to 0} 1 + 2h \sin\big(\frac{1}{h}\big) = 1 \tag{390}$$
Therefore,
$$f'(t) = \begin{cases} 1 + 4t\sin\left(\frac{1}{t}\right) - 2\cos\left(\frac{1}{t}\right) & \text{if } x \neq 0 \\ 1 & \text{if } x = 0 \end{cases} \tag{391}$$
And on $t \in (-1,1)$ we see that f' is clearly bounded. To prove that it is not injective for any neighborhood of 0, it suffices to construct a sequence of points t_n such that the derivatives oscillate around 0.[a] Consider $t_n = \frac{1}{\pi n}$, which converges to 0. Then, we have the nice property that $\sin(\frac{1}{t_n}) = 0$ and
$$\cos\left(\frac{1}{t_n}\right) = \cos(\pi n) = (-1)^n \tag{392}$$
Therefore, the derivative is
$$f'(t_n) = 1 + 4t_n \sin\left(\frac{1}{t_n}\right) - 2\cos\left(\frac{1}{t_n}\right) = 1 - 2\cos(\pi n) = 1 - 2\cdot(-1)^n \tag{393}$$
Therefore, for every open interval $(-\epsilon, \epsilon)$ around 0, we can find an $n \in \mathbb{N}$ so large that the point t_n with derivative $f'(t_n) = 3$ and t_{n+1} with derivative $f'(t_{n+1}) = -1$. Since f' changes signs within this interval, f cannot be monotone, and since it cannot be monotone, f cannot be injective.

[a] Note that since the derivative is $1 + 4t\sin\left(\frac{1}{t}\right) - 2\cos\left(\frac{1}{t}\right)$.

Exercise 1.12 (Rudin 9.17)

Let $f = (f_1, f_2)$ be the mapping of $\mathbb{R}^2$ into $\mathbb{R}^2$ given by
$$f_1(x,y) = e^x \cos y, \qquad f_2(x,y) = e^x \sin y. \tag{394}$$
(a) What is the range of f?

(b) Show that the Jacobian of f is not zero at any point of $\mathbb{R}^2$. Thus every point of $\mathbb{R}^2$ has a neighborhood in which f is one-to-one. Nevertheless, f is not one-to-one on $\mathbb{R}^2$.

(c) Put

$$a = \left(0, \frac{\pi}{3}\right), \qquad b = f(a), \tag{395}$$

let g be the continuous inverse of f, defined in a neighborhood of b, such that $g(b) = a$. Find an explicit formula for g, compute $f'(a)$ and $g'(b)$, and verify the formula (52).

(d) What are the images under f of lines parallel to the coordinate axes?

Solution. Listed.

1. $\text{range}(f) = \mathbb{R}^2 \setminus \{0\}$.
2. Compute
$$Df(x,y) = \begin{pmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} \end{pmatrix} = \begin{pmatrix} e^x \cos(y) & -e^x \sin(y) \\ e^x \sin(y) & e^x \cos(y) \end{pmatrix} \neq 0 \tag{396}$$
for all $x, y \in \mathbb{R}^2$ since $\sin(y) = 0 \implies \cos(y) \neq 0$. f is also clearly continuous. So, by the inverse function theorem, f is locally invertaible, but f not globally injective since $f(x,y) = f(x, y+2\pi)$.
3. So,
$$a = \begin{pmatrix} 0 \\ \frac{\pi}{3} \end{pmatrix}, \qquad b = f(a) = \begin{pmatrix} \frac{1}{2} \\ \frac{\sqrt{3}}{2} \end{pmatrix} \tag{397}$$
We can write
$$g \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \ln(\sqrt{x^2+y^2}) \\ \arctan(y/x) \end{pmatrix} = \begin{pmatrix} \frac{1}{2}\ln(x^2+y^2) \\ \arctan(y/x) \end{pmatrix} \tag{398}$$
So, the derivative is
$$Dg(x,y) = \begin{pmatrix} \frac{x}{x^2+y^2} & \frac{y}{x^2+y^2} \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix} \tag{399}$$
Evaluating gives
$$Dg(\frac{1}{2}, \frac{\sqrt{3}}{2}) = \begin{pmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}^{-1} = Df(0, \frac{\pi}{3})^{-1} \tag{400}$$
4. Circles and lines(?).

Exercise 1.13 (Rudin 9.18)

Answer analogous questions for the mapping defined by

$$u = x^2 - y^2, \qquad v = 2xy. \tag{401}$$

Solution. Listed.

1. $\text{range}(f) = \mathbb{R}^2(?)$.
2. We see that
$$Df(x,y) = \begin{pmatrix} 2x & -2y \\ 2y & 2x \end{pmatrix} \tag{402}$$
So $Df \neq 0$ if $(x,y) \neq 0$. f is also continuous, so f is locally invertible. However, f is not globally invertaible since $f(x,y) = f(-x,-y)$.

3. To invert this, we can see that

$$\begin{cases} u^2 &= (x^2 - y^2)^2 = x^4 - 2x^2y^2 + y^4 \\ v^2 &= (2xy)^2 = 4x^2y^2 \end{cases} \implies \sqrt{u^2+v^2} = x^2 + y^2 \tag{403}$$

Therefore,

$$\sqrt{\frac{\sqrt{u^2+v^2}+u}{2}} = \sqrt{\frac{x^2+y^2+x^2-y^2}{2}} = \sqrt{x^2} = \text{sign}(x)|x| \tag{404}$$

$$\sqrt{\frac{\sqrt{u^2+v^2}-u}{2}} = \sqrt{\frac{x^2+y^2-x^2+y^2}{2}} = \sqrt{y^2} = \text{sign}(y)|y| \tag{405}$$

So,

$$g(u,v) = \left(\sqrt{\frac{\sqrt{u^2+v^2}+u}{2}}, \sqrt{\frac{\sqrt{u^2+v^2}-u}{2}}\right) \tag{406}$$

After some tedious computations, we get

$$Dg(x,y) = \begin{pmatrix} \frac{1}{4}\left(\frac{\sqrt{u^2+v^2}+u}{2}\right)^{-\frac{1}{2}}\left(\frac{u}{\sqrt{u^2+v^2}}+1\right) & \frac{1}{4}\left(\frac{\sqrt{u^2+v^2}+u}{2}\right)^{-\frac{1}{2}}\left(\frac{v}{\sqrt{u^2+v^2}}+1\right) \\ \frac{1}{4}\left(\frac{\sqrt{u^2+v^2}-u}{2}\right)^{-\frac{1}{2}}\left(\frac{u}{\sqrt{u^2+v^2}}-1\right) & \frac{1}{4}\left(\frac{\sqrt{u^2+v^2}-u}{2}\right)^{-\frac{1}{2}}\left(\frac{v}{\sqrt{u^2+v^2}}-1\right) \end{pmatrix} \tag{407}$$

Therefore,

$$Dg(0,2) = \begin{pmatrix} \frac{1}{4} & \frac{1}{4} \\ -\frac{1}{4} & \frac{1}{4} \end{pmatrix} = \begin{pmatrix} 2 & -2 \\ 2 & 2 \end{pmatrix}^{-1} = Df(1,1)^{-1} \tag{408}$$

4. The images are the parabolas opening towards the origin.

Exercise 1.14 (Rudin 9.19)

Show that the system of equations

$$3x + y - z + u^2 = 0, \tag{409}$$

$$x - y + 2z + u = 0, \tag{410}$$

$$2x + 2y - 3z + 2u = 0 \tag{411}$$

can be solved for x, y, u in terms of z; for x, z, u in terms of y; for y, z, u in terms of x; but not for x, y, z in terms of u.

Solution. This is basically a straightforward application of the implicit function theorem. We have $F : \mathbb{R}^{1+3} \to \mathbb{R}^3$, with $n = 1, m = 3$.

$$F\begin{pmatrix} x \\ y \\ z \\ w \end{pmatrix} = \begin{pmatrix} 3x + y - z + u^2 \\ x - y + 2z + u \\ 2x + 2y - 3z + 2u \end{pmatrix} \tag{412}$$

Now, we take the (partial) Jacobian of F to get

$$D_{(x,y,z)}F(x,y,z) = \begin{pmatrix} 3 & 1 & -1 \\ 1 & -1 & 2 \\ 2 & 2 & -3 \end{pmatrix} \implies |D_{(x,y,z)}F(x,y,z)| = 0 \tag{413}$$

$$D_{(x,y,u)}F(x,y,u) = \begin{pmatrix} 3 & 1 & 2u \\ 1 & -1 & 1 \\ 2 & 2 & 2 \end{pmatrix} \implies |D_{(x,y,u)}F(x,y,u)| = 8u - 12 \tag{414}$$

$$D_{(x,z,u)}F(x,z,u) = \begin{pmatrix} 3 & -1 & 2u \\ 1 & 2 & 1 \\ 2 & -3 & 2 \end{pmatrix} \implies |D_{(x,z,u)}F(x,z,u)| = -14u + 2 \tag{415}$$

$$D_{(y,z,u)}F(y,z,u) = \begin{pmatrix} 1 & -1 & 2u \\ -1 & 2 & 1 \\ 2 & -3 & 2 \end{pmatrix} \implies |D_{(y,z,u)}F(y,z,u)| = -2u + 3 \tag{416}$$

So by the implicit function theorem, the claim holds.

Exercise 1.15 (Rudin 9.20)

Take $n = m = 1$ in the implicit function theorem, and interpret the theorem (as well as its proof) graphically.

Solution. Basically means that if the graph in $\mathbb{R}^2$ is continuous and the tangent line is not vertical, we can locally represent y as function of x.

Exercise 1.16 (Rudin 9.21)

Define f in $\mathbb{R}^2$ by

$$f(x,y) = 2x^3 - 3x^2 + 2y^3 + 3y^2. \tag{417}$$

(a) Find the four points in $\mathbb{R}^2$ at which the gradient of f is zero. Show that f has exactly one local maximum and one local minimum in $\mathbb{R}^2$.

(b) Let S be the set of all $(x,y) \in \mathbb{R}^2$ at which $f(x,y) = 0$. Find those points of S that have no neighborhoods in which the equation $f(x,y) = 0$ can be solved for y in terms of x (or for x in terms of y). Describe S as precisely as you can.

Solution. Listed.

1. We have

$$\nabla f(x,y) = \begin{pmatrix} 6x^2 - 6x \\ 6y^2 + 6y \end{pmatrix} \tag{418}$$

The gradient equals 0 if

$$\begin{pmatrix} x \\ y \end{pmatrix} \in \left\{ \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ -1 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right\} \tag{419}$$

We can determine if this is a local extrema or a saddle point with the second derivative test. Since

$$Hf(x,y) = \begin{pmatrix} 12x - 6 & 0 \\ 0 & 12y + 6 \end{pmatrix}, \tag{420}$$

we have

$$Hf(0,0) = \begin{pmatrix} -6 & 0 \\ 0 & 6 \end{pmatrix} \implies \text{saddle point} \tag{421}$$

$$Hf(1,0) = \begin{pmatrix} 6 & 0 \\ 0 & 6 \end{pmatrix} \implies \text{local minimum} \tag{422}$$

$$Hf(0,-1) = \begin{pmatrix} -6 & 0 \\ 0 & -6 \end{pmatrix} \implies \text{local maximum} \tag{423}$$

$$Hf(1,-1) = \begin{pmatrix} 6 & 0 \\ 0 & -6 \end{pmatrix} \implies \text{saddle point} \tag{424}$$

2. By the (contrapositive of the) implicit function theorem, if a point cannot be locally solved for y in terms of x, then the derivative $D_y f(x,y) = 0$. Similarly, if a point cannot be locally solved for x in terms of y, then $D_x f(x,y) = 0$. Since

$$Df(x,y) = \begin{pmatrix} 6x^2 - 6x & 6y^2 + 6y \end{pmatrix} = \begin{pmatrix} 0 & 0 \end{pmatrix} \tag{425}$$

So,

(a) For there to be no representation of y as a function of x, we must have $6y^2 + 6y = 0 \implies y = 0, -1$. In the first case when $y = 0$, $(x, 0)$ must lie on S, so

$$f(x,0) = 2x^3 - 3x^2 = 0 \implies x = 0, \frac{3}{2} \tag{426}$$

In the second case when $y = -1$, $(x,-1)$ must lie on S, so

$$f(x,-1) = 2x^3 - 3x^2 + 1 = 0 \implies x = 1, -\frac{1}{2} \tag{427}$$

Therefore, the 4 desired points are

$$(0,0), (\frac{3}{2}, 0), (1,-1), (-\frac{1}{2}, -1) \tag{428}$$

(b) For there to be no representation of x as a function of y, we must have $6x^2 - 6x = 0 \implies x = 0, 1$. In the first case when $x = 0$, $(0, y)$ must lie on S, so

$$f(0,y) = 2y^3 + 3y^2 = 0 \implies y = 0, -\frac{3}{2} \tag{429}$$

In the second case when $x = 1$, $(1, y)$ must lie on S, so

$$f(1,y) = 2y^3 + 3y^2 - 1 = 0 \implies y = -1, \frac{1}{2} \tag{430}$$

Therefore, the 4 desired points are

$$(0,0), (0, -\frac{3}{2}), (1,-1), (1, \frac{1}{2}) \tag{431}$$

Exercise 1.17 (Rudin 9.22)

Give a similar discussion for

$$f(x,y) = 2x^3 + 6xy^2 - 3x^2 + 3y^2. \tag{432}$$

Solution. The gradient is

$$\nabla f(x,y) = \begin{pmatrix} 6x^2 + 6y^2 - 6x \\ 12xy + 6y \end{pmatrix} \tag{433}$$

For the gradient to equal 0, first $12xy + 6y = 6y(2x+1) = 0$. This means that $y = 0$ or $x = -\frac{1}{2}$. When $y = 0$, this means that $6x^2 - 6x = 0 \implies x = 0, 1$. When $x = -\frac{1}{2}$, this means that $6y^2 + \frac{9}{2} = 0 \implies y^2 = -\frac{3}{4}$, which has no solution. Therefore, $\nabla f(x,y) = 0$ only at $(0,0)$ or $(1,0)$. If we look at the Hessian, we get

$$Hf(0,0) = \begin{pmatrix} -6 & 0 \\ 0 & 6 \end{pmatrix} \implies \text{saddle point}, \quad Hf(1,0) = \begin{pmatrix} 6 & 0 \\ 0 & 18 \end{pmatrix} \implies \text{local minimum} \tag{434}$$

For there to be no local representation of y as a function of x, it must be the case that $D_y f(x,y) = 12xy + 6y = 0$, so $y = 0$ or $x = -\frac{1}{2}$.

1. In the first case when $y = 0$, $(x, 0)$ must lie on S, so $f(x,0) = 2x^3 - 3x^2 = 0 \implies x = 0, \frac{3}{2}$.
2. In the second case when $x = -\frac{1}{2}$, $(-\frac{1}{2}, y)$ must lie on S, so $f(-\frac{1}{2}, y) = -1 = 0$, which implies that there is no solution.

So only points are $(0,0)$ and $(\frac{3}{2}, 0)$. The same logic can be applied for finding points that have no local representation of x as a function of y.

Exercise 1.18 (Rudin 9.23)

Define f in $\mathbb{R}^3$ by

$$f(x, y_1, y_2) = x^2 y_1 + e^x + y_2. \tag{435}$$

Show that $f(0,1,-1) = 0$, $(D_1 f)(0,1,-1) \neq 0$, and that there exists therefore a differentiable function g in some neighborhood of $(1,-1)$ in $\mathbb{R}^2$, such that $g(1,-1) = 0$ and

$$f(g(y_1,y_2), y_1, y_2) = 0. \tag{436}$$

Find $(D_1 g)(1,-1)$ and $(D_2 g)(1,-1)$.

Solution. Clearly, $f(0,1,-1) = 0$ and

$$\frac{\partial f}{\partial x} = 2xy_1 + e^x \implies \frac{\partial f}{\partial x}(0,1,-1) = 1 \neq 0 \tag{437}$$

We wish to write x as a function of y_1, y_2, i.e. $x = x(y_1, y_2)$. By the implicit function theorem, there exists a neighborhood U of $(1,-1) \in \mathbb{R}^2$ s.t. $g(-1,1) = 0$ and

$$f\big(g(y_1,y_2), y_1, y_2\big) = 0 \tag{438}$$

We know that

$$Df(x, y_1, y_2) = \big(\underbrace{2xy_1 + e^x}_{D_x f}, \underbrace{x^2, 1}_{D_y f}\big) \tag{439}$$

and $Df(0,1,-1) = (1,0,1)$. So by the implicit function theorem,

$$\begin{aligned} Dg(1,-1) &= -\big(D_x f(1,0,1)\big)^{-1}\big(D_y f(1,0,1)\big) && (440)\\ &= -(1)^{-1}(0,1) && (441)\\ &= (0,-1) && (442) \end{aligned}$$

So, $\frac{\partial g}{\partial y_1} = 0$, $\frac{\partial g}{\partial y_2} = -1$.

Exercise 1.19 (Rudin 9.24)

For $(x, y) \neq (0, 0)$, define $f = (f_1, f_2)$ by

$$f_1(x, y) = \frac{x^2 - y^2}{x^2 + y^2}, \qquad f_2(x, y) = \frac{xy}{x^2 + y^2}. \tag{443}$$

Compute the rank of $f'(x, y)$, and find the range of f.

Solution. This is just very tedious calculations. We can compute that

$$Df(x, y) = \frac{1}{(x^2 + y^2)^2} \begin{pmatrix} 4xy^2 & -4x^2 y \\ -x^2 y + y^3 & x^3 - xy^2 \end{pmatrix} \implies |Df(x, y)| = 0 \tag{444}$$

Since $Df(x, y)$ is a nontrivial matrix, the rank must be 1. To find the range, we can see that

$$\big(f_1(x, y)\big)^2 + \big(2f_2(x, y)\big)^2 = \frac{1}{(x^2 + y^2)^2}(x^4 + 2x^2 y^2 y^4) = 1 \tag{445}$$

So the range must lie in an ellipse.

Exercise 1.20 (Rudin 9.25)

Suppose $A \in L(\mathbb{R}^n, \mathbb{R}^m)$, and let r be the rank of A.

(a) Define S as in the proof of Theorem 9.32. Show that SA is a projection in $\mathbb{R}^n$ whose null space is $\mathscr{N}(A)$ and whose range is $\mathscr{R}(S)$.
Hint: By (68), $SASA = SA$.

(b) Use (a) to show that

$$\dim \mathscr{N}(A) + \dim \mathscr{R}(A) = n. \tag{446}$$

Exercise 1.21 (Rudin 9.26)

Show that the existence (and even the continuity) of $D_{12}f$ does not imply the existence of $D_1 f$. For example, let $f(x, y) = g(x)$, where g is nowhere differentiable.

Solution. Let

$$f(x, y) = \begin{cases} 1 & \text{if } x \in \mathbb{Q} \\ 0 & \text{if } x \notin \mathbb{Q} \end{cases} \tag{447}$$

Then, $\frac{\partial f}{\partial y} = 0$ and so $\frac{\partial^2 f}{\partial x \partial y} = 0$, which is continuous. However, $\frac{\partial f}{\partial x}$ does not exist everywhere.

Exercise 1.22 (Rudin 9.27)

Put $f(0, 0) = 0$, and

$$f(x, y) = \frac{xy(x^2 - y^2)}{x^2 + y^2} \tag{448}$$

if $(x, y) \neq (0, 0)$. Prove that

(a) f, $D_1 f$, and $D_2 f$ are continuous in $\mathbb{R}^2$;

(b) $D_{12} f$ and $D_{21} f$ exist at every point of $\mathbb{R}^2$, and are continuous except at $(0, 0)$;

(c)

$$(D_{12} f)(0, 0) = 1, \qquad (D_{21} f)(0, 0) = -1. \tag{449}$$

Solution. Listed.

1. *Continuity of f.* In $\mathbb{R}^2 \setminus \{0\}$, it is clearly continuous. At 0, we have $|x^2 - y^2| \leq |x^2 + y^2|$, so

$$|f(x,y)| = \frac{|xy(x^2 - y^2)|}{x^2 + y^2} \leq |xy| \implies 0 \leq \lim_{(x,y)\to(0,0)} |f(x,y)| \leq \lim_{(x,y)\to(0,0)} |xy| = 0 \tag{450}$$

so f is continuous at 0.

2. *Partials are continuous.* In $\mathbb{R}^2 \setminus \{0\}$, we can use the quotient rule to find

$$\frac{\partial f}{\partial x} = \frac{\partial}{\partial x} \frac{x^3 y - xy^3}{x^2 + y^2} = \frac{(3x^2 y - y^3)(x^2 + y^2) - (x^3 y - xy^3)(2x)}{(x^2 + y^2)^2} = \frac{y(x^2 - y^2)}{x^2 + y^2} \tag{451}$$

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y} \frac{x^3 y - xy^3}{x^2 + y^2} = \frac{(x^3 - 3xy^2)(x^2 + y^2) - (x^3 y - xy^3)(2y)}{(x^2 + y^2)^2} = \frac{x(x^2 - y^2)}{x^2 + y^2} \tag{452}$$

which are continuous. At the origin, we have

$$\frac{\partial f}{\partial x}(0,0) = \lim_{h \to 0} \frac{f(h,0) - f(0,0)}{h} = \lim_{h\to 0} \frac{0 - 0}{h} = 0 \tag{453}$$

$$\frac{\partial f}{\partial y}(0,0) = \lim_{h \to 0} \frac{f(0,h) - f(0,0)}{h} = \lim_{h\to 0} \frac{0 - 0}{h} = 0 \tag{454}$$

To see that the partials are continuous at the origin, we can see that

$$\lim_{(x,y)\to(0,0)} \left| \frac{\partial f}{\partial x}(x,y) \right| = \lim_{(x,y)\to(0,0)} \left| \frac{y(x^2 - y^2)}{x^2 + y^2} \right| \leq \lim_{(x,y)\to(0,0)} |y| = 0 \tag{455}$$

$$\lim_{(x,y)\to(0,0)} \left| \frac{\partial f}{\partial y}(x,y) \right| = \lim_{(x,y)\to(0,0)} \left| \frac{x(x^2 - y^2)}{x^2 + y^2} \right| \leq \lim_{(x,y)\to(0,0)} |x| = 0 \tag{456}$$

Therefore, the limits are both 0, equal to the values of the partial derivatives at $(0,0)$.

3. *Second Partials.* We can compute

$$\frac{\partial^2 f}{\partial x \partial y}(0,0) = \lim_{h \to 0} \frac{1}{h} \left(\frac{\partial f}{\partial y}(h,0) - \frac{\partial f}{\partial y}(0,0) \right) \tag{457}$$

$$= \lim_{h\to 0} \frac{1}{h} \frac{h^3}{h^2} = 1 \tag{458}$$

$$\frac{\partial^2 f}{\partial y \partial x}(0,0) = \lim_{h \to 0} \frac{1}{h} \left(\frac{\partial f}{\partial x}(0,h) - \frac{\partial f}{\partial x}(0,0) \right) \tag{459}$$

$$= \lim_{h\to 0} \frac{1}{h} \frac{-h^3}{h^2} = -1 \tag{460}$$

Exercise 1.23 (Rudin 9.28)

For $t \geq 0$, put

$$\varphi(x,t) = \begin{cases} x, & 0 \leq x \leq \sqrt{t}, \\ -x + 2\sqrt{t}, & \sqrt{t} \leq x \leq 2\sqrt{t}, \\ 0, & \text{otherwise}, \end{cases} \tag{461}$$

and put $\varphi(x,t) = -\varphi(x,|t|)$ if $t < 0$.
Show that φ is continuous on $\mathbb{R}^2$, and

$$(D_2 \varphi)(x,0) = 0 \tag{462}$$

for all x. Define
$$f(t) = \int_{-1}^{1} \varphi(x,t)\,dx. \tag{463}$$
Show that $f(t) = t$ if $|t| < \frac{1}{4}$. Hence
$$f'(0) \neq \int_{-1}^{1} (D_2\varphi)(x,0)\,dx. \tag{464}$$

Solution. Listed.

1. *Continuous.* If we look at the "overlaps," we can see that φ agrees on the boundaries of the piecewise domains, so I thought of using the pasting lemma. Consider the closed sets
$$A = \{(x,t) \in \mathbb{R}^2 \colon 0 \leq t, 0 \leq x \leq \sqrt{t}\} \tag{465}$$
$$B = \{(x,t) \in \mathbb{R}^2 \colon 0 \leq t, \sqrt{t} \leq x \leq 2\sqrt{t}\} \tag{466}$$
$$C = \{(x,t) \in \mathbb{R}^2 \colon 0 \leq t, 2\sqrt{t} \leq x\} \tag{467}$$
Consider the continuous functions
$$\varphi_A : A \to \mathbb{R}, \quad \varphi(x,t) = x \tag{468}$$
$$\varphi_B : B \to \mathbb{R}, \quad \varphi(x,t) = -x + 2\sqrt{t} \tag{469}$$
$$\varphi_C : C \to \mathbb{R}, \quad \varphi(x,t) = 0 \tag{470}$$
Since φ_A, φ_B agree on $A \cap B$, and φ_B, φ_C agree on $B \cap C$, by the pasting lemma, we can extend the continuous functions to a continuous function φ (and do the same for the domain where $t < 0$). This leads to the function:
$$\varphi(x,t) = \begin{cases} x & \text{if } 0 \leq t, 0 \leq x \leq \sqrt{t} \\ -x + 2\sqrt{t} & \text{if } 0 \leq t, \sqrt{t} \leq x \leq 2\sqrt{t} \\ 0 & \text{if } 0 \leq t, \text{ else} \\ -x & \text{if } t < 0, 0 \leq x \leq \sqrt{-t} \\ x - 2\sqrt{-t} & \text{if } t < 0, \sqrt{-t} \leq x \leq 2\sqrt{-t} \\ 0 & \text{if } t < 0, \text{ else} \end{cases} \tag{471}$$

2. *Partial Derivative.* Let x be fixed, and by definition
$$\frac{\partial \varphi}{\partial t}(x,0) = \lim_{h \to 0} \frac{\varphi(x,h) - \varphi(x,0)}{h} = \lim_{h \to 0} \frac{\varphi(x,h)}{h} \tag{472}$$
Since φ is defined piecewise for when $h \geq 0$ and $h < 0$, we split it into left and right hand limits.
   (a) When $h > 0$, we have
$$\varphi(x,h) = \begin{cases} x & \text{if } 0 \leq x \leq \sqrt{h} \\ -x + 2\sqrt{h} & \text{if } \sqrt{h} \leq x \leq 2\sqrt{h} \\ 0 & \text{if } 2\sqrt{h} < x \end{cases} = \begin{cases} x & \text{if } x^2 \leq h \\ -x + 2\sqrt{h} & \text{if } \frac{x^2}{4} \leq h \leq x^2 \\ 0 & \text{if } 0 < h < \frac{x^2}{4} \end{cases} \tag{473}$$
   and $\varphi(0,h) = 0$.
   (b) We also have $-h < 0$, so
$$\varphi(x,-h) = \begin{cases} x & \text{if } 0 \leq x \leq \sqrt{-h} \\ -x + 2\sqrt{-h} & \text{if } \sqrt{-h} \leq x \leq 2\sqrt{-h} \\ 0 & \text{if } 2\sqrt{-h} < x \end{cases} = \begin{cases} x & \text{if } h \leq -x^2 \\ -x + 2\sqrt{-h} & \text{if } -x^2 < h < -\frac{x^2}{4} \\ 0 & \text{if } -\frac{x^2}{4} < h < 0 \end{cases} \tag{474}$$

Since $\varphi(x,0) = 0$, we know that for every $x \in \mathbb{R}$, there exists a neighborhood around $t = 0$ where φ is constantly 0 (and so dividing it by h still gives 0), which implies that $\lim_{h\to 0} \frac{\varphi(x,h)}{h} = \lim_{h\to 0} \frac{0}{h} = 0$. Therefore, the partial derivative evaluates to 0 for every x.

3. If $|t| < \frac{1}{4}$, we have $\sqrt{|t|} < \frac{1}{2} \implies 2\sqrt{|t|} < 1$. When $t = 0$, $\varphi(x,t) = 0 \implies \int_{-1}^1 \varphi(x,t)\,dx = 0$. When $t > 0$,

$$f(t) = \int_{-1}^1 \varphi(x,t)\,dx \tag{475}$$
$$= \int_{-1}^0 \underbrace{\varphi(x,t)}_{=0}\,dx + \int_0^{\sqrt{t}} \varphi(x,t)\,dx + \int_{\sqrt{t}}^{2\sqrt{t}} \varphi(x,t)\,dx + \int_{2\sqrt{t}}^1 \underbrace{\varphi(x,t)}_{=0}\,dx \tag{476}$$
$$= \int_0^{\sqrt{t}} x\,dx + \int_{\sqrt{t}}^{2\sqrt{t}} -x + 2\sqrt{t}\,dx = t \tag{477}$$

When $t < 0$,

$$f(t) = \int_{-1}^1 \varphi(x,t)\,dx \tag{478}$$
$$= \int_{-1}^0 \underbrace{\varphi(x,t)}_{=0}\,dx + \int_0^{\sqrt{-t}} \varphi(x,t)\,dx + \int_{\sqrt{-t}}^{2\sqrt{-t}} \varphi(x,t)\,dx + \int_{2\sqrt{-t}}^1 \underbrace{\varphi(x,t)}_{=0}\,dx \tag{479}$$
$$= \int_0^{\sqrt{-t}} -x\,dx + \int_{\sqrt{-t}}^{2\sqrt{-t}} x - 2\sqrt{-t}\,dx = t \tag{480}$$

Hence,

$$f'(0) = \left(\frac{d}{dt}\left[\int_{-1}^1 \varphi(x,t)\,dx\right]\right)(0) = \left(\frac{d}{dt} t\right)(0) = 1 \neq 0 = \frac{\partial \varphi}{\partial t}(x,0) \tag{481}$$

Exercise 1.24 (Rudin 9.29)

Let E be an open set in $\mathbb{R}^n$. The classes $\mathscr{C}^{(k)}(E)$ and $\mathscr{C}^{(\infty)}(E)$ are defined in the text. By induction, $\mathscr{C}^{(k)}(E)$ can be defined as follows, for all positive integers k: To say that $f \in \mathscr{C}^{(k)}(E)$ means that the partial derivatives $D_1 f, \ldots, D_n f$ belong to $\mathscr{C}^{(k-1)}(E)$.
Assume $f \in \mathscr{C}^{(k)}(E)$, and show (by repeated application of Theorem 9.41) that the kth-order derivative

$$D_{i_1 i_2 \cdots i_k} f = D_{i_1} D_{i_2} \cdots D_{i_k} f \tag{482}$$

is unchanged if the subscripts $i_1, \ldots, i_k$ are permuted.
For instance, if $n \geq 3$, then

$$D_{1213} f = D_{3112} f \tag{483}$$

for every $f \in \mathscr{C}^{(4)}$.

Exercise 1.25 (Rudin 9.30)

Let $f \in \mathscr{C}^{(m)}(E)$, where E is an open subset of $\mathbb{R}^n$. Fix $a \in E$, and suppose $x \in \mathbb{R}^n$ is so close to 0 that the points

$$p(t) = a + tx \tag{484}$$

lie in E whenever $0 \leq t \leq 1$. Define

$$h(t) = f(p(t)) \tag{485}$$

for all $t \in \mathbb{R}^1$ for which $p(t) \in E$.

(a) For $1 \leq k \leq m$, show (by repeated application of the chain rule) that

$$h^{(k)}(t) = \sum (D_{i_1 \cdots i_k} f)(p(t)) x_{i_1} \cdots x_{i_k}. \tag{486}$$

The sum extends over all ordered k-tuples $(i_1, \ldots, i_k)$ in which each i_j is one of the integers $1, \ldots, n$.

(b) By Taylor's theorem (5.15),

$$h(1) = \sum_{k=0}^{m-1} \frac{h^{(k)}(0)}{k!} + \frac{h^{(m)}(t)}{m!} \tag{487}$$

for some $t \in (0, 1)$. Use this to prove Taylor's theorem in n variables by showing that the formula

$$f(a + x) = \sum_{k=0}^{m-1} \frac{1}{k!} \sum (D_{i_1 \cdots i_k} f)(a) x_{i_1} \cdots x_{i_k} + r(x) \tag{488}$$

represents $f(a + x)$ as the sum of its so-called "Taylor polynomial of degree $m - 1$," plus a remainder that satisfies

$$\lim_{x \to 0} \frac{r(x)}{|x|^{m-1}} = 0. \tag{489}$$

Each of the inner sums extends over all ordered k-tuples $(i_1, \ldots, i_k)$, as in part (a); as usual, the zero-order derivative of f is simply f, so that the constant term of the Taylor polynomial of f at a is $f(a)$.

(c) Exercise 29 shows that repetition occurs in the Taylor polynomial written in part (b). For instance, D_{113} occurs three times, as D_{113}, D_{131}, and D_{311}. The sum of the corresponding three terms can be written in the form

$$3(D_1^2 D_3 f)(a) x_1^2 x_3. \tag{490}$$

Prove, by calculating how often each derivative occurs, that the Taylor polynomial in (b) can be written in the form

$$\sum \frac{(D_1^{s_1} \cdots D_n^{s_n} f)(a)}{s_1! \cdots s_n!} x_1^{s_1} \cdots x_n^{s_n}. \tag{491}$$

Here the summation extends over all ordered n-tuples $(s_1, \ldots, s_n)$ such that each s_i is a nonnegative integer, and

$$s_1 + \cdots + s_n \leq m - 1. \tag{492}$$

Exercise 1.26 (Rudin 9.31)

Suppose $f \in \mathscr{C}^{(3)}$ in some neighborhood of a point $a \in \mathbb{R}^2$, the gradient of f is 0 at a, but not all second-order derivatives of f are 0 at a. Show how one can then determine from the Taylor polynomial of f at a (of degree 2) whether f has a local maximum, or a local minimum, or neither, at the point a. Extend this to $\mathbb{R}^n$ in place of $\mathbb{R}^2$.